CSE 429 · COMPUTER VISION AND PATTERN RECOGNITION · MIDTERM REFERENCE

# Seven lectures, six sheets. Here's the short road through them.

For the midterm on **Tuesday 10 November 2026, 13:30** · Scope: Lectures 1–7 and Sheets 1–6 · Gonzalez & Woods, *Digital Image Processing* (3rd ed.), Chapters 1–3 · written for someone starting from zero.

## Is it doable by tomorrow?

**Yes, if you follow the core path and don't chase every slide.** Most of the marks are methods: count a histogram and equalize it, find the shortest m-path, stretch a 3 × 3 image, run a filter over a small matrix, take a derivative of an array. Each one is taught here on a small problem first, then the real sheet problem is boxed for you to solve, with the full answer at the end of the lecture.

All seven lectures add up to about **5 hours**. That is deliberately lean: slides that are pictures or history are reduced to a few lines, and every sheet question is kept. Keep about **5½ hours of sleep**: a rested brain does arithmetic faster and with fewer slips, and this exam is arithmetic.

# A suggested plan (exam Tuesday 13:30)

| When | What | ⏱ |
|---|---|---|
| 22:45 | Lecture 1: what vision is | 10 |
| 22:55 | Lecture 6: spatial filtering + Sheet 5 Q3–Q5, Sheet 6 Q1–Q6 | 55 |
| 23:50 | Lecture 2: digital images + Sheets 1 and 3 | 30 |
| 00:20 | Lecture 3, part one: pixels, paths, distances + Sheet 2 | 35 |
| 00:55 | Lecture 3, part two: interpolation, operations + Sheet 4 Q1–Q3 | 25 |
| *01:25* | *Sleep* | |
| 07:00 | Lecture 4: stretching, slicing, equalization | 35 |
| 07:35 | Lecture 5: histogram matching + Sheet 4 Q4–Q6, Sheet 5 Q1–Q2 | 45 |
| 08:20 | Lecture 7, part 1: derivatives and the Laplacian (Sheet 6 Q7–Q10) | 30 |
| 12:30 | Lecture 7, part 2 (Q11–Q13) + the formula sheet from memory | 40 |

Lecture 6 comes early on purpose: it stands on its own, and it carries the most sheet questions.

# How to read this document

**📖 Read & understand**
No pen. Read it once slowly, then say the idea back to yourself in one sentence.

**✏️ Pen & paper**
The method is taught on a smaller problem first. Then the real sheet problem is boxed for you to solve on paper. Its full answer is at the end of the lecture.

**🔁 Recall drill**
Quick questions. Answer before you look.

Tier A must do · Tier B should do · Tier C only if you're ahead. Minutes (⏱) are for a tired student, not a fresh one. Each lecture opens with an "Only 20 minutes?" path.

**Links.** ↗ Understand more links open the interactive study guide (`vision-study.html`, same folder) or a short outside explanation. Use them when a card didn't click.

🧭 **Analogies** tie each idea to something familiar and say where the comparison stops working.

**Mistakes in the course material** are flagged in each lecture's ⚠️ Traps box, with the correct version.

# Contents


1. **Lecture 1** · What computer vision is, and why it's hard · 4 cards · ⏱ 10 min
2. **Lecture 2** · Digital image fundamentals: from light to numbers · 6 cards · ⏱ 30 min
3. **Lecture 3 (pp. 1–19)** · Pixel relationships: neighbours, adjacency, paths, regions, distances · 5 cards · ⏱ 35 min
4. **Lecture 3 (pp. 20–42)** · Interpolation and operations on whole images · 5 cards · ⏱ 25 min
5. **Lecture 4** · Image enhancement: point transforms and histograms · 5 cards · ⏱ 35 min
6. **Lecture 5** · Histogram processing: equalization and matching · 7 cards · ⏱ 45 min
7. **Lecture 6** · Spatial filtering: correlation, convolution, smoothing · 7 cards · ⏱ 55 min
8. **Lecture 7** · Sharpening: derivatives, the Laplacian, unsharp masking, Sobel · 10 cards · ⏱ 53 min

LECTURE 1

# What computer vision is, and why it's hard

Tier B · should ⏱ 10 min Slides: Lec 1 - Overview of Computer Vision and Pattern Recognition.pdf (72 p, mostly pictures)

📘 Guide Tutorial 1

Lecture 1 is the map of the course: what the field tries to do, how its problems are sorted into levels, and why a task a toddler does without effort is still research. There are no calculations here, only definitions and examples, so it is quick marks if you read it once.

**What the exam can ask from this lecture**

- Computer vision vs image processing vs computer graphics, in one line each.
- The three levels of vision and an example task at each level.
- Why vision is ill-posed, and the named challenges (viewpoint, illumination, occlusion…).
- An example application for a named area (biometrics, automotive safety, medicine…).

⏳ **Only 5 minutes?** Read card 2, then do the drill.

📖 READ & UNDERSTAND · ⏱ 3 min · Tier B

## 1 · Three fields that all touch images

Take one photo of a street with a car, a horse and a pole. **Image processing** makes the photo better for a person to look at: sharper, brighter, less noisy. The output is still an image.

**Computer vision** has to say what is in it: "a car, a horse, a person; the car is in front of the pole". The output is a description that a computer acts on. **Computer graphics** runs the other way: it starts from a model and creates the picture.

Between them sits **image analysis**, which has two parts: **feature extraction** (turning pixels into higher-level information, such as edges or shapes) and **pattern classification** (using that information to name objects). The course title, *Computer Vision and Pattern Recognition*, is those two halves.

🧭 **Like a phone camera app.** The beauty filter that smooths your skin is image processing; the face-unlock that decides "this is the owner" is computer vision; the AR sticker drawn on your face is graphics. *Where it breaks:* real apps chain all three, so one product rarely belongs to only one field.

↗ Understand more: guide Tutorial 1, step 1

📖 READ & UNDERSTAND · ⏱ 3 min · Tier B

## 2 · Low, mid and high: sorting vision tasks by how much they understand

| Level | Tasks (Lec 1) | On the street photo |
|---|---|---|
| Low | measurements, enhancement, region segmentation, features | the white patch, the shadow, the edges |
| Mid | reconstruction, depth, motion estimation | the car is about 1 m in front of the pole |
| High | category detection, activity recognition, pose, deep understanding | a car, a horse, a man riding past |

Lecture 2 uses a second, similar list from the textbook, about **processes**: low-level (noise reduction, contrast, sharpening; image in, image out), mid-level (segmentation, classifying objects; image in, attributes out), high-level (making sense of the scene). Notice that segmentation sits at **low** level in Lecture 1's vision list and at **mid** level in Lecture 2's processing list. Answer with whichever list the question names.

🧭 **Like reading a page.** Low level is seeing the ink strokes, mid level is grouping them into words and noticing the page is tilted, high level is understanding the story. *Where it breaks:* in vision the levels feed back into each other; knowing "that's a car" helps find its edges.

↗ Understand more: guide Tutorial 1, step 2

📖 READ & UNDERSTAND · ⏱ 2 min · Tier C

## 3 · Why a summer project became a whole field

In 1966 Marvin Minsky asked an undergraduate to connect a camera to a computer over one summer and make it "describe what it sees". It is still research. The core reason: a camera flattens a 3-D world into a 2-D picture and throws depth away, so **many different scenes give exactly the same image**. Recovering the scene from the image has no unique answer without extra assumptions; that is what **ill-posed** means.

Lecture 1 names nine practical challenges. The pictures make them easy to remember:

| | |
|---|---|
| viewpoint | one statue from three angles |
| illumination | one face, lit two ways |
| occlusion | a rider half-hidden by trees |
| scale | a tall and a short man holding laptops |
| deformation | galloping horses |
| background clutter | a figure dissolving into a patterned painting |
| intra-class variation | six very different chairs, all "chair" |
| local ambiguity | a blurry patch that only makes sense in context |
| the world behind the image | knowing a train is long and heavy without seeing it |

🧭 **Like a shadow puppet.** A hand, a cardboard cut-out and a real dog can all cast the same shadow; from the shadow alone you can't be sure which it was. *Where it breaks:* a camera keeps colour and texture too, which is why vision is hard but not hopeless.

↗ Understand more: guide Tutorial 1, steps 3–4

🔁 RECALL DRILL · ⏱ 2 min · Tier B

## 4 · Quick recall

1. A system estimates how fast a car is moving between two video frames. Which level of vision?
2. A dog recogniser works in daylight and fails at dusk. Which challenge?
3. Six photos of different chairs are all labelled "chair". Which challenge?
4. In one line: why is vision called ill-posed?
5. Creating images from a 3-D model is which field?

✍️ Answers: S1.1 (end of this lecture)

### ⚠️ Traps

- Segmentation is "low level" in Lecture 1's vision list but a "mid-level process" in Lecture 2's list. Use the list the question asks about.
- Image processing outputs an image for people; computer vision outputs a description for a machine. Don't swap them.

# Solutions and answers — Lecture 1

### S1.1 · 4 · Quick recall — Answers

1. Mid level (motion estimation).
2. Illumination.
3. Intra-class variation.
4. Many 3-D scenes project to the same 2-D image, so the answer isn't unique.
5. Computer graphics.

↩ back to the card

LECTURE 2

# Digital image fundamentals: from light to numbers

Tier A · must ⏱ 30 min Slides: Lec 2-Digital Image Fundamentals.pdf (63 p) · Sheet 1 (16 MCQs), Sheet 3 Q1–Q6

↗ Szeliski Ch 1–2

Before any filter or histogram can touch a picture, the picture has to become a grid of numbers. This lecture is that conversion: how a camera samples and quantizes a scene, how many bits the result needs, and how long it takes to send. Everything after it assumes you can picture an image as $f(x, y)$, a table of integers.

**What the exam can ask from this lecture**

- Image size in bits, bytes and MB for a given resolution, number of levels and grey/RGB (Lec 2 p37–40, Sheet 3 Q1–Q2).
- Transmission time over a modem with start/stop bits (Sheet 3 Q3–Q4); HDTV movie storage (Q5–Q6).
- Definitions: computer vision vs image processing vs graphics, low/mid/high level, sampling vs quantization, spatial vs intensity resolution, dynamic range vs contrast.
- MCQs in the style of Sheet 1, mostly on the key stages.

⏳ **Only 20 minutes?** Do cards 3, 6 and 2, in that order.

📖 READ & UNDERSTAND · ⏱ 4 min · Tier A

## 1 · Who is the picture for: a person or a computer?

The slides split the field by one question: **who consumes the output** (L2 p4). In **image processing** the output is an image for a human to look at (a brighter X-ray, a sharper photo). In **computer vision** the output is used by a computer ("a car at (120, 40)", "this is digit 7"). **Computer graphics** runs the other way: it *creates* images from a model (L2 p6).

Work is also sorted into three **levels** by what goes in and comes out (L2 p5):

| level | examples | input → output |
|---|---|---|
| low | noise reduction, contrast enhancement, sharpening | image → image |
| mid | segmentation, classifying individual objects | image → attributes (regions, labels) |
| high | image analysis, the "understanding" of human vision | attributes → meaning |

Quick test: sharpening returns an image, so it is **low-level**; cutting a photo into "road" and "sky" returns regions, so it is **mid-level**. Within computer vision, **image analysis** = feature extraction (pull edges, shapes out of pixels) + pattern classification (name the objects from those features, L2 p8).

🧭 **Like a phone camera app.** The beauty filter that smooths your skin is image processing (you look at it); the face detector that draws the focus box is computer vision (the phone uses it); the cartoon sticker is graphics. *Where it breaks:* real apps chain all three; the exam wants the textbook split.

More: Szeliski Ch 1 (szeliski.org/Book); the guide's Chapter 2 (vision-study.html).

📖 READ & UNDERSTAND · ⏱ 5 min · Tier A

## 2 · Two roundings turn a scene into a table

A sensor's output is a **continuous waveform**: any position, any value (L2 p30). The slides model an image as a **2-D function** $f(x, y)$: $(x, y)$ is a position and $f$ the **intensity** (grey level) there (L2 p14). It is a **digital image** once $x$, $y$ *and* $f$ are all discrete; each cell is a **pixel**, and an $M \times N$ image has $MN$ of them (L2 p15–17).

Two separate steps get you there (L2 p30). **Sampling** digitizes the **coordinates**: where to measure. **Quantization** digitizes the **amplitude**: each value is rounded to one of $L$ levels. Tiny run, 3 bits so $L = 8$ (levels 0–7), rule level $=$ round$(7v)$:

| sample $v$ | 0.12 | 0.47 | 0.83 | 0.95 | 0.50 |
|---|---|---|---|---|---|
| $7v$ | 0.84 | 3.29 | 5.81 | 6.65 | 3.5 |
| level | 1 | 3 | 6 | 7 | 4 |

Each kind of rounding has its own resolution. **Spatial resolution** = smallest visible detail, in line pairs or dots per unit distance (**dpi**); you need the dpi *and* the print size, not just the pixel count (L2 p34). **Intensity resolution** = smallest visible change in intensity, in **bits per pixel**, since $k$ bits give $L = 2^k$ levels (L2 p35): 1 bit → 2, 3 → 8, 8 → 256, 9 → 512, 16 → 65,536. Too few levels breaks smooth shading into bands, **false contouring** (the skulls on L2 p36).

Don't mix up two look-alikes (L2 p33): **dynamic range** is a **ratio**, max measurable ÷ min detectable intensity (a property of the system); **contrast** is a **difference**, highest − lowest intensity in this image (values 40…200 → contrast 160).

🧭 **Like copying a photo as a mosaic.** Sampling decides how many tiles there are; quantization decides how many shades of tile are in your box, so each tile gets the nearest shade you own. *Where it breaks:* a real sensor averages light over each cell instead of reading one point.

Szeliski Ch 2.3 covers the sensor side: szeliski.org/Book.

🖍 PEN & PAPER · ⏱ 9 min · Tier A

## 3 · How many bits, and how long to send them

> IN PLAIN WORDS
> ① Sheet 3 Q1–Q2 give a resolution (rows × columns), the number of intensity levels $L$, and grey (1 channel) or RGB (3 channels, red/green/blue); find the size. Q3–Q4 send an image over a modem whose **baud rate** is bits per second, in **packets** of start bit + data + stop bit; find the time.

> WARM-UP BY HAND
> ② A 100×50 RGB image with 16 levels: $\log_2 16 = 4$ bits × 3 channels = 12 bits per pixel; $5000 \times 12 = 60{,}000$ bits $= 7{,}500$ bytes. Sent in 8-bit packets, that is 7,500 packets × 10 line bits = 75,000 bits; at 9,600 baud, $75{,}000/9{,}600 = 7.81$ s.

> THE ONE IDEA

**③ Size (bits) = rows × columns × channels × $\log_2 L$. On the wire, every packet costs 2 extra bits, so count packets first, then multiply by (data bits + 2).**

BUILD IT

④ on the slides' camera (L2 p37–40): 4 MP = 2680 × 1520, RGB, 256 levels.

1. Pixels: $2680 \times 1520 = 4{,}073{,}600$.
   Why: one pixel per grid cell.
2. Bits per pixel: $\log_2 256 = 8$ per channel × 3 = 24.
   Why: RGB stores one 8-bit value per colour.
3. $4{,}073{,}600 \times 24 = 97{,}766{,}400$ bits; ÷ 8 = 12,220,800 bytes.
   Why: 1 byte = 8 bits.
4. ÷ 1,048,576 = **11.655 MB**.
   Why: the slides take 1 MB = $2^{20}$ bytes.

CHECK

⑤ Dividing by $10^6$ instead gives 12.22 MB: wrong unit, not wrong maths. Forgetting the 3 channels gives a third of the answer. More: the guide's Chapter 2, Steps 4–5 (vision-study.html).

EXAM PROBLEM

**Your turn — Sheet 3 Q1.** A camera with resolution 4 Mega Pixel (2680 × 1520 pixel). What is the size of the output RGB image if it has 512 intensity levels?
**Q2.** A camera with resolution 2 Mega Pixel (1920 × 1080 pixel). What is the size of the output grey image if it has 128 intensity levels?
**Q3.** Transmission is in packets of a start bit, a byte (8 bits) of information and a stop bit. (a) How many minutes to transmit a 1024 × 1024 image with 256 intensity levels using a 56K baud modem? (b) The time at 3000K baud (a medium-speed DSL line)?
**Q4.** Now packets carry a start bit, 2 bytes (16 bits) of information and a stop bit. (a) How many minutes for a 1024 × 1024 image with 65,536 intensity levels at 70K baud? (b) The time at 4000K baud and 256 intensity levels?

✍ Show the full exam answer: S2.1 (end of this lecture)

TRY ALONE

⑦ A 512 × 512 grey image with 64 levels: size in bits and in MB?

✍ Answers: S2.2 (end of this lecture)

🖍 PEN & PAPER · ⏱ 5 min · Tier B

## 4 · A movie is just frames × bits per frame

IN PLAIN WORDS

① Sheet 3 Q5–Q6 fix the number of horizontal TV **lines** (the image height in pixels) and the **aspect ratio** width:height; pixels per line = lines × ratio. Each pixel has 24 bits (8 each for R, G, B). Find the bits for a movie of given length at 30 frames per second.

**WARM-UP BY HAND**

② 90 lines at 16:9 → $90 \times 16/9 = 160$ pixels per line. One frame = $160 \times 90 \times 24 = 345{,}600$ bits; one second at 30 fps = 10,368,000 bits.

**THE ONE IDEA**

③ **Total bits = (lines × lines·ratio) × 24 × fps × seconds.** "Interlaced" (two half-fields of 1/60 s) changes nothing: two fields make one frame and the question already gives frames per second.

**BUILD IT**

④ : 720 lines, 4:3, 24 bits, a 10-second clip at 25 fps.

1. Width = $720 \times 4/3 = 960$ pixels per line.
   Why: width/height = 4/3 and height = 720.
2. Per frame = $960 \times 720 \times 24 = 16{,}588{,}800$ bits.
   Why: pixels × bits per pixel.
3. Frames = $25 \times 10 = 250$.
   Why: fps × seconds.
4. Total = $16{,}588{,}800 \times 250 = 4{,}147{,}200{,}000$ bits.
   Why: every frame is stored in full (no compression in these questions).

**CHECK**

⑤ Hours become seconds: 2 h = 7,200 s, not 120. If the width isn't whole (1000 × 4/3), keep the fraction and say so. More: the guide's Chapter 2, Step 6 (vision-study.html).

**EXAM PROBLEM**

**Your turn — Sheet 3 Q5.** HDTV with 1125 horizontal lines, interlaced (two fields of 1/60 s each), aspect ratio 16:9. Pixels per line are in proportion to the vertical resolution, the proportion being the width-to-height ratio. Each pixel has 24 bits (8 per R, G, B). How many bits to store a 2-hour HDTV movie at 30 frames per second?

**Q6.** The same with 1000 lines, aspect ratio 4:3, and a 1-hour movie at 30 frames per second.

✍ Show the full exam answer: S2.3 (end of this lecture)

**TRY ALONE**

⑦ 1080 lines at 16:9, 24 bits, one minute at 30 fps: how many bits?

✍ Answers: S2.4 (end of this lecture)

📖 READ & UNDERSTAND · ⏱ 3 min · Tier B

## 5 · The rest of the lecture in five short facts

**Key stages** (L2 p19), in the slide's arrow order: acquisition → enhancement → restoration → morphological processing → segmentation → object recognition → representation & description, with colour processing and compression underneath, used wherever needed. One line each: **enhancement**

makes it look better (subjective); **restoration** undoes a known degradation such as noise or blur (objective, model-based); **segmentation** partitions the image into regions; **recognition** attaches a label to an object; **compression** shrinks storage.

**Image sources** (L2 p9–13): EM spectrum, acoustic, ultrasonic, electron beams, synthetic. Gamma rays: PET, where the isotope emits **positrons** that annihilate with electrons and the detectors catch the resulting **gamma rays**. X-rays: CT (CAT). Satellite imagery: GIS.

**Simultaneous contrast** (L2 p27): equal grey squares look darker on lighter backgrounds, so perceived brightness does not depend *simply* on intensity. Like the viral "what colour is the dress" photo: same pixels, different perception.

**skimage code** (L2 p41–61): `io.imread` gives a uint8 array 0–255 of shape rows × columns × 3; `img_as_float` rescales to 0.0–1.0 (÷255), `img_as_ubyte` back to 0–255; `img*0.3` on the float image darkens it; a 2-D array needs `plt.imshow(a, 'gray')` or it shows in false colour. Slicing is `img[rows, cols, channel]`: `img[:50, :, :] = [255, 0, 0]` paints the top 50 rows red, `img[:, :50, :]` is the left 50 columns.

All five with figures and tap-through code: the guide's Chapters 2–3 (vision-study.html).

RECALL DRILL · ⏱ 4 min · Tier A

## 6 · Sheet 1 in one pass

Cover the answers and say yours aloud; the questions are shortened, the options are the sheet's.

1. Digital image processing is: (A) computer algorithms on digital images (B) analog → digital (C) physical filters (D) printing.
2. The order of the key stages: options A–D (C follows acquisition → enhancement → restoration → morphology → segmentation → recognition → representation → compression → colour).
3. NOT an acquisition device: (A) cameras (B) range sensors (C) tomography (D) image enhancement techniques.
4. Purpose of enhancement: (A) produce images with sensors (B) improve visual quality (C) go to the frequency domain (D) operate on the Fourier transform.
5. Enhancement working directly on pixels: (A) spatial domain (B) transform (C) Fourier (D) frequency.
6. Works on the Fourier transform and back: (A) spatial (B) transform domain (C) frequency domain (D) pixel-based.
7. Goal of restoration: (A) enhance quality (B) remove noise from the image (C) segment (D) extract shape components.
8. Common causes of noise: (A) camera shake and low light (B) motion blur and mis-focus (C) corrupted files, pixelation (D) over/under-exposure.
9. Edge detection belongs to: (A) enhancement (B) restoration (C) morphology (D) segmentation.
10. Goal of segmentation: (C) partition the image into sets of pixels, or another option?
11. Local vs global segmentation: (A) local uses small windows on the image (B) global has fewer pixels (C) edges vs noise (D) large vs small images.
12. Primary goal of segmentation: (a) complicate (b) discard information (c) simplify/change the representation for easier analysis (d) complicate analysis.

13. Object recognition: (c) perceive an object's properties and attach semantic attributes, or another?
14. Representing an object by its boundary: (a) Huffman (b) run-length (c) lossy compression (d) global segmentation.
15. Lossy compression: (a) exact reconstruction (b) accepts losing some information (c) visible spectrum (d) enhances details.
16. What segmentation aims to achieve (same options as Q12).

✍️ Answers: S2.5 (end of this lecture)

### ⚠️ Traps

- L2 p10 says the isotope "emits positrons (Gamma rays)" and calls a bone scan a PET example. A positron is not a gamma ray: the detectors see the two gamma rays from positron–electron annihilation, and a bone scan is plain gamma-ray imaging, not PET.
- L2 p27 says perceived brightness "does not depend on its intensity". Correct version: it does not depend *simply* on intensity; the background changes it too.
- L2 p33 attaches dynamic range to "an image"; it is really a property of the imaging system (a ratio), while contrast is a difference within one image.
- Levels → bits is $\log_2 L$: 256 levels is 8 bits, not 256. RGB multiplies by 3; grey doesn't.
- MB in these slides = $2^{20}$ bytes (11.655 MB on L2 p40; the slide's last line rounds it to 11.66). Dividing by $10^6$ gives 12.22.
- Transmission: each packet adds a start and a stop bit, so a byte costs 10 line bits (18 for a 16-bit packet). Sheet 3 Q4(b) is ambiguous: two 8-bit pixels per packet gives 2.36 s, one per packet 4.72 s.
- uint8 arithmetic wraps: in NumPy, uint8 250 + 10 = 4, not 260 or 255. Convert with `img_as_float` before doing maths.
- Sheet 1: Q14 has no correct option, Q6 has two defensible ones, Q8 mixes blur with noise, Q16 repeats Q12.

## Solutions and answers — Lecture 2

### S2.1 · 3 · How many bits, and how long to send them — Show the full exam answer

**Q1.** $512 = 2^9 \rightarrow 9$ bits per channel × 3 = 27 bits per pixel. Size $= 2680 \times 1520 \times 27 = 4{,}073{,}600 \times 27 = 109{,}987{,}200$ bits $= 13{,}748{,}400$ bytes $= 13{,}748{,}400/1{,}048{,}576 \approx 13.11$ MB.

**Q2.** Grey = 1 channel; $128 = 2^7 \rightarrow 7$ bits. Size $= 1920 \times 1080 \times 7 = 2{,}073{,}600 \times 7 = 14{,}515{,}200$ bits $= 1{,}814{,}400$ bytes $\approx 1.73$ MB.

**Q3.** 256 levels → 8 bits = one byte = one packet per pixel. Packet = $1 + 8 + 1 = 10$ bits. Total $= 1024 \times 1024 \times 10 = 10{,}485{,}760$ bits.
(a) $10{,}485{,}760/56{,}000 = 187.2$ s $= 187.2/60 \approx 3.12$ min. (b) $10{,}485{,}760/3{,}000{,}000 \approx 3.50$ s.

**Q4.** Packet $= 1 + 16 + 1 = 18$ bits.
(a) 65,536 levels → 16 bits = one packet per pixel: $1{,}048{,}576 \times 18 = 18{,}874{,}368$ bits; $/70{,}000 = 269.6$ s $\approx 4.49$ min.
(b) 256 levels → 8 bits, so two pixels fill one 16-bit packet: $1{,}048{,}576/2 = 524{,}288$ packets × 18 = 9,437,184 bits; $/4{,}000{,}000 \approx 2.36$ s. (If you assume one pixel per packet, half the data bits wasted: $1{,}048{,}576 \times 18/4{,}000{,}000 \approx 4.72$ s. State your assumption.)

**Q1 109,987,200 bits ≈ 13.11 MB · Q2 14,515,200 bits ≈ 1.73 MB · Q3 (a) ≈ 3.12 min (b) ≈ 3.50 s · Q4 (a) ≈ 4.49 min (b) ≈ 2.36 s (4.72 s with one pixel per packet)**

↩ back to the card

### S2.2 · 3 · How many bits, and how long to send them — Answers

$\log_2 64 = 6$; $262{,}144 \times 6 = 1{,}572{,}864$ bits $= 196{,}608$ bytes $= 0.1875$ MB.

↩ back to the card

### S2.3 · 4 · A movie is just frames × bits per frame — Show the full exam answer

**Q5.** Vertical resolution = 1125 lines. Pixels per line $= 1125 \times 16/9 = 2000$. Bits per frame $= 2000 \times 1125 \times 24 = 54{,}000{,}000$. Frames $= 30 \times 2 \times 3600 = 216{,}000$. Total $= 54{,}000{,}000 \times 216{,}000 = 1.1664 \times 10^{13}$ bits.

**Q6.** Pixels per line $= 1000 \times 4/3 = 1333.\overline{3}$. Bits per frame $= 1000 \times \frac{4000}{3} \times 24 = 32{,}000{,}000$. Frames $= 30 \times 3600 = 108{,}000$. Total $= 32{,}000{,}000 \times 108{,}000 = 3.456 \times 10^{12}$ bits. (Rounding to 1333 whole pixels per line gives $\approx 3.455 \times 10^{12}$; say which you used.)

**Q5:** $1.1664 \times 10^{13}$ **bits · Q6:** $3.456 \times 10^{12}$ **bits**

↩ back to the card

### S2.4 · 4 · A movie is just frames × bits per frame — Answers

$1080 \times 16/9 = 1920$; $1920 \times 1080 \times 24 = 49{,}766{,}400$ bits per frame × 1,800 frames $= 89{,}579{,}520{,}000 \approx 8.96 \times 10^{10}$ bits.

↩ back to the card

### S2.5 · 6 · Sheet 1 in one pass — Answers

1. A.
2. C (it matches the arrows on L2 p19; colour and compression aren't really in the chain).
3. D: enhancement is processing, not acquisition.
4. B.
5. A: spatial domain, $g(x, y) = T[f(x, y)]$.
6. B and C are both defensible: frequency domain is a kind of transform domain (Lecture 4's word is "transform"). Flawed question.
7. B (restoration undoes degradations; A is enhancement).
8. A is the intended answer, but only "low light" causes noise; camera shake causes blur. Ambiguous question.
9. D: edge detection is a segmentation tool.
10. C.
11. A: local works on small windows; global uses the whole image (more pixels, so B is backwards).
12. c.
13. c.
14. None is correct: all four are compression or segmentation. Boundary representation uses chain codes or polygonal approximation (Gonzalez Ch 11).
15. b.
16. c: the same question as Q12.

↩ back to the card

LECTURE 3 (PP. 1–19)

# Pixel relationships: neighbours, adjacency, paths, regions, distances

Tier A · must ⏱ 35 min Slides: Lec 3 pp. 1–19 · Sheet 2 Q1–Q4 ↗ Szeliski Ch 3

Lecture 2 turned a picture into a grid of numbers. This part asks which numbers **belong together**: which pixels touch, whether you can walk from one to another through pixels of the right value, and how far apart two pixels are. Every later idea that talks about "an object" or "a region" (and the 3×3 neighbourhoods of filtering in Lecture 6) stands on these definitions.

**What the exam can ask from this lecture**

- Define $N_4$, $N_D$, $N_8$, and 4-, 8- and m-adjacency for a value set V.
- Decide whether two pixels, or two image subsets, are 4-, 8- or m-adjacent.
- Find (draw) the shortest 4-, 8- and m-paths between p and q and give their lengths, or explain why a path does not exist.
- Count the connected components (regions) of a binary image under 4-, 8- and m-adjacency.
- Compute the Euclidean, city-block ($D_4$) and chessboard ($D_8$) distances between two pixels.

⏳ **Only 20 minutes?** Do cards 1 and 2, then the drill (card 5).

📖 READ & UNDERSTAND · ⏱ 9 min · Tier A

## 1 · Who touches whom: neighbours, adjacency, paths, regions

Put your finger on one pixel $p$ at row $x$, column $y$. Four pixels share an edge with it: $(x \pm 1, y)$ and $(x, y \pm 1)$. They are its **4-neighbours**, $N_4(p)$. Four more touch it only at a corner, $(x \pm 1, y \pm 1)$: the **diagonal neighbours** $N_D(p)$. Together they make the **8-neighbours**, $N_8(p) = N_4(p) \cup N_D(p)$ (L3 p2, p5). A pixel on the image border has fewer, because some of them fall outside.

| | | |
|---|---|---|
| | × | |
| × | p | × |
| | × | |

| | | |
|---|---|---|
| × | | × |
| | p | |
| × | | × |

| | | |
|---|---|---|
| × | × | × |
| × | p | × |
| × | × | × |

Neighbours are about position only. **Adjacency** adds a condition on the values: pick a set $V$ of allowed grey levels (for a binary image $V = \{1\}$; $V = \{2, 3\}$ means "only pixels whose value is 2 or 3 count"). Two pixels $p, q$ *both with values in* $V$ are (L3 p5):

- **4-adjacent** if $q \in N_4(p)$;
- **8-adjacent** if $q \in N_8(p)$;
- **m-adjacent** (mixed) if $q \in N_4(p)$, *or* $q \in N_D(p)$ and $N_4(p) \cap N_4(q)$ contains no pixel with a value in $V$.

That last line reads badly but works simply. A diagonal pair sits in a 2×2 square; the other two cells of that square are $N_4(p) \cap N_4(q)$, the **two corner pixels**. If either corner is in $V$, refuse the diagonal (you can already walk round through that corner). If both corners are outside $V$, the diagonal is allowed. Slide toy (L3 p6), $V = \{1\}$:

| | | |
|---|---|---|
| 0 | 1 | 1 |
| 0 | 1 | 0 |
| 0 | 0 | 1 |

The top-right 1 and the centre 1 are diagonal; their corners are the 1 on the top row and the 0 on the right, and that 1 is in $V$, so under m the diagonal is refused (8-adjacency would allow it, giving two routes between the same pixels). The bottom-right 1 and the centre are diagonal with corners 0 and 0, so that diagonal is kept. **m-adjacency exists to remove the double routes that 8-adjacency creates.**

🧭 **Like a chess king (8) versus a rook moving one square (4).** m is a king that may only step diagonally when it can't go round by two straight steps. *Where it breaks:* a king may stand on any square; here every square on the route must hold a value in $V$.

A **path** from $p$ to $q$ is a chain of pixels, each adjacent to the next (4-, 8- or m-path depending on the adjacency used). Its **length** is the number of **steps**, i.e. pixels − 1: a path through 5 pixels has length 4. Two pixels are **connected** in a set $S$ if a path inside $S$ joins them, and a **connected component** (region) is a largest group of mutually connected pixels (L3 p11). The slide's 5×4 binary image has three 1-blocks touching only at corners: **3 regions** under 4-adjacency, **1** under 8. m always gives the same *regions* as 8 (it only drops a diagonal when the corner already connects the two pixels); it changes the *paths*.

Background reading: Gonzalez & Woods §2.5, or the interactive guide's Tutorial 4 (vision-study.html), which lights up every path and region in this section.

✏️ **PEN & PAPER · ⏱ 13 min · Tier A**

## 2 · Shortest 4-, 8- and m-paths, and when none exists

**IN PLAIN WORDS**

① Given an image, a value set $V$, a start pixel $p$ and an end pixel $q$, find the fewest steps from $p$ to $q$ when every pixel on the way has a value in $V$ and each step is a 4-, 8- or m-adjacency. Coordinates here are (row, column) counted from 1, top-left.

**WARM-UP BY HAND**

② $V = \{1\}$, $p$ bottom-left, $q$ top-right of $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$. 4-path: up, right, up, right = **4**. 8-path: two diagonals through the centre = **2**. m-path: both diagonals have a 1 in a corner, so they are refused and m falls back to the 4-route = **4**.

The slides do the same on bigger grids. L3 p7–p9, $V = \{2, 3\}$: no 4-path, shortest 8-path 3 (straight up the diagonal), the only m-path has length 5. Figure 5 (L3 p10, $V = \{2, 3, 5\}$): no 4-path, 8-path 4, m-path 5.

THE ONE IDEA

**③ Walk outward from $p$ one step at a time and stop the first time you reach $q$; for every diagonal step, check its two corner pixels before you take it under m.**
A 4-path is impossible as soon as $p$ or $q$ has no 4-neighbour in $V$.

BUILD IT

④ (the warm-up, m-path).

1. Check the ends: $p$'s 4-neighbour above is a 1, so $p$ is not stuck.
   Why: if an end has no neighbour in $V$, stop and write "no path".
2. Try the diagonal $p \rightarrow$ centre: corners are the 1 above $p$ and the 0 to its right. A corner is in $V \rightarrow$ refused.
   Why: m rule from card 1.
3. Go round: $p \rightarrow (2,1) \rightarrow (2,2)$, then centre $\rightarrow q$ is refused the same way, so $(2,2) \rightarrow (1,2) \rightarrow (1,3)$.
   Why: the refused diagonal is replaced by its two straight steps.
4. Count steps: 4 (five pixels).
   Why: length = pixels − 1.

CHECK

⑤ Always 8-length ≤ m-length, and m-length ≤ 4-length whenever a 4-path exists: each extra rule can only make the route longer. The guide's Tutorial 4 (vision-study.html) has an explorer that runs these searches on any grid.

EXAM PROBLEM

**Your turn — Sheet 2 Q2.** Consider the image segment shown. (a) Let $V = \{0, 1\}$ and compute the lengths of the shortest 4-, 8- and m-path between $p$ and $q$. If a particular path does not exist between these two points, explain why. (b) Repeat for $V = \{1, 2\}$.

| | | | |
|---|---|---|---|
| 3 | 1 | 2 | 1 (q) |
| 2 | 2 | 0 | 2 |
| 1 | 2 | 1 | 1 |
| (p) 1 | 0 | 1 | 2 |

**Sheet 2 Q3.** According to the following intensity values of a 5×5 image, show the paths by drawing dashed lines for 4-adjacency, 8-adjacency and m-adjacency. Let $V = \{1, 2, 3\}$. From $p$ to $q$.

| | | | | |
|---|---|---|---|---|
| 5 | 8 | 3 | 4 | 2 (q) |
| 4 | 8 | 7 | 1 | 3 |
| 3 | 6 | 1 | 8 | 3 |
| 8 | 2 | 3 | 6 | 1 |
| (p) 2 | 4 | 6 | 9 | 3 |

✍ Show the full exam answer: S3.1 (end of this lecture)

TRY ALONE

⑦ (a) A path visits 7 pixels; its length? (b) A diagonal pair's corners hold 0 and 4, $V = \{2, 4\}$: m-adjacent? (c) Could an m-path ever be shorter than the 8-path between the same pixels?

✍ Answers: S3.2 (end of this lecture)

✎ PEN & PAPER · ⏱ 7 min · Tier A

## 3 · Adjacent subsets and counting regions

IN PLAIN WORDS

① Two **subsets** $S_1, S_2$ of an image are adjacent if *some* pixel of $S_1$ is adjacent to *some* pixel of $S_2$; you only need to look along their shared border. Counting regions means grouping all pixels in $V$ into connected components and counting the groups.

WARM-UP BY HAND

② $\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$, $V = \{1\}$. Under 4-adjacency the right-hand pair of 1s is one region and the three diagonal 1s are alone: **4 regions**. Under 8 (and m) the diagonals join everything: **1 region**.

THE ONE IDEA

③ **Edge contact joins under all three adjacencies; corner-only contact joins under 8 and m but never under 4.** So for regions, compare only 4 against 8; m always matches 8.

BUILD IT

④ on the same warm-up grid, split into subsets $S_1$ = columns 1–2, $S_2$ = columns 3–4.

1. 1s next to the border (columns 2 and 3): $(2, 2)$ in $S_1$, $(3, 3)$ in $S_2$.
   Why: only pixels next to the border can be adjacent across it.
2. $(2, 2)$ and $(3, 3)$ meet at a corner: not 4-adjacent, but 8-adjacent.
   Why: diagonal contact.
3. Corners $(2, 3) = 0$, $(3, 2) = 0$: both outside $V$, so also m-adjacent.
   Why: the m rule allows a diagonal with two empty corners.

CHECK

⑤ Across a shared border, 8-adjacent subsets are always m-adjacent too: a corner pixel in $V$ would itself lie in $S_1$ or $S_2$ and be 4-adjacent across the border. The guide's Tutorial 4 (vision-study.html) colours each region of Q4 when you tap a pixel.

EXAM PROBLEM

**Your turn — Sheet 2 Q1.** Consider the two image subsets $S_1$ and $S_2$ in the figure. For $V = \{1\}$, determine whether these two subsets are (a) 4-adjacent, (b) 8-adjacent, or (c) m-adjacent. ($S_1$ = rows 1–4, columns 2–5; $S_2$ = rows 1–4, columns 6–9; the dashed boxes on the sheet.)

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 |

**Sheet 2 Q4.** Find the regions in the binary matrix below, where $V = \{1\}$ [4-adjacency, 8-adjacency and m-adjacency].

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

✍ Show the full exam answer: S3.3 (end of this lecture)

**TRY ALONE**

⑦ (a) Two 1s touch only at a corner and both corners are 0: same 4-region? same m-region? (b) Can going from 4- to 8-adjacency ever increase the number of regions?

✍ Answers: S3.4 (end of this lecture)

📖 **READ & UNDERSTAND · ⏱ 3 min · Tier A**

## 4 · Three ways to measure distance between pixels

How far is $q = (s, t)$ from $p = (x, y)$? It depends on how you are allowed to move (L3 p13–p16):

| name | formula | moves like | pixels with $D \le r$ form |
|---|---|---|---|
| Euclidean $D_e$ | $\sqrt{(x-s)^2 + (y-t)^2}$ | a straight line | a disc |
| city-block $D_4$ | $\lvert x-s\rvert + \lvert y-t\rvert$ | a taxi on a street grid (4-steps) | a diamond (13 pixels for $r = 2$) |
| chessboard $D_8$ | $\max(\lvert x-s\rvert, \lvert y-t\rvert)$ | a chess king (8-steps) | a square (5×5 for $r = 2$) |

Sibling: $p = (2, 5)$, $q = (6, 2)$, so the row and column gaps are 4 and 3. $D_4 = 4 + 3 = 7$, $D_8 = \max(4, 3) = 4$, $D_e = \sqrt{16 + 9} = 5$. **Always** $D_8 \le D_e \le D_4$.

The slides' examples: L3 p17, $q = (1, 1)$, $p = (2, 2)$: $D_e = \sqrt{2} \approx 1.41$, $D_4 = 2$, $D_8 = 1$. L3 p19, $p = (4, 2)$, $q = (1, 4)$: $D_4 = \lvert 1-4\rvert + \lvert 2-4\rvert = 3 + 2 = 5$, $D_8 = \max(3, 2) = 3$, $D_e = \sqrt{9 + 4} = \sqrt{13} \approx 3.6$. These ignore pixel values; the length of a 4- or m-path (card 2) depends on them, so it can be longer than $D_4$ or not exist at all.

The guide's Tutorial 4 (vision-study.html) draws all three distance maps around a pixel.

🔁 RECALL DRILL · ⏱ 3 min · Tier A

## 5 · Quick recall

1. How many pixels are in $N_8(p)$ for an interior pixel? In $N_4(p)$ for a corner pixel of the image?
2. State the m-adjacency rule for a diagonal pair in one sentence.
3. A path passes through 9 pixels. Length?
4. $p = (3, 3)$, $q = (1, 6)$: $D_4$, $D_8$, $D_e$?
5. Why do 8- and m-adjacency always give the same number of regions?

✍️ Answers: S3.5 (end of this lecture)

### ⚠️ Traps

- Path length counts steps, not pixels: 5 pixels = length 4.
- Under m, check the two corner pixels of *every* diagonal step; a corner value in $V$ refuses the diagonal (Sheet 2 Q2 and Q3 both have refused diagonals).
- "No 4-path" needs a reason: name the end pixel whose 4-neighbours are all outside $V$ (Sheet 2 Q2(a): $q$; Q3: $p$).
- m changes paths, never regions: Sheet 2 Q4 gives 3 regions under both 8 and m.
- L3 p19 redraws the 4×4 example with $p$ at $(4, 2)$ (row 4 printed as 2 2 0 1), whereas L3 p7 puts $p$ at $(4, 1)$ (row 4 = 2 0 0 1). Use the coordinates each slide marks; the distances only depend on the coordinates.

## Solutions and answers — Lecture 3 (pp. 1–19)

### S3.1 · 2 · Shortest 4-, 8- and m-paths, and when none exists — Show the full exam answer

**Q2 (a)** $V = \{0, 1\}$**.** $p = (4, 1)$, $q = (1, 4)$.

**4-path: does not exist.** The 4-neighbours of $q$ are $(1, 3) = 2$ and $(2, 4) = 2$, both outside $V$, so no 4-step can ever arrive at $q$.

**8-path: length 4.** $(4, 1) \to (4, 2) \to (3, 3) \to (2, 3) \to (1, 4)$, values 1, 0, 1, 0, 1.

| | | | |
|---|---|---|---|
| 3 | 1 | 2 | $1_q$ |
| 2 | 2 | 0 | 2 |
| 1 | 2 | 1 | 1 |
| $1_p$ | 0 | 1 | 2 |

**m-path: length 5.** The diagonal $(4, 2) \to (3, 3)$ has corners $(3, 2) = 2$ and $(4, 3) = 1$; the 1 is in $V$, so it is refused and the path goes through $(4, 3)$. The last diagonal $(2, 3) \to (1, 4)$ has corners $(1, 3) = 2$, $(2, 4) = 2$, both outside $V$: allowed. Path $(4, 1) \to (4, 2) \to (4, 3) \to (3, 3) \to (2, 3) \to (1, 4)$.

| | | | |
|---|---|---|---|
| 3 | 1 | 2 | $1_q$ |
| 2 | 2 | 0 | 2 |
| 1 | 2 | 1 | 1 |
| $1_p$ | 0 | 1 | 2 |

**Q2 (b)** $V = \{1, 2\}$**.**

**4-path: length 6.** $(4, 1) \to (3, 1) \to (2, 1) \to (2, 2) \to (1, 2) \to (1, 3) \to (1, 4)$ (values 1, 1, 2, 2, 1, 2, 1). The 0s at $(4, 2)$ and $(2, 3)$ block the shorter routes.

**8-path: length 4.** $(4, 1) \to (3, 1) \to (2, 2) \to (1, 3) \to (1, 4)$.

| | | | |
|---|---|---|---|
| 3 | 1 | 2 | $1_q$ |
| 2 | 2 | 0 | 2 |
| 1 | 2 | 1 | 1 |
| $1_p$ | 0 | 1 | 2 |

**m-path: length 6.** Both diagonals of the 8-path are refused: $(3, 1) \to (2, 2)$ has corner $(2, 1) = 2 \in V$, and $(2, 2) \to (1, 3)$ has corner $(1, 2) = 1 \in V$. Going round them gives the same cells as the 4-path:

| | | | |
|---|---|---|---|
| 3 | 1 | 2 | $1_q$ |
| 2 | 2 | 0 | 2 |
| 1 | 2 | 1 | 1 |
| $1_p$ | 0 | 1 | 2 |

**Q2 (a)** $V = \{0, 1\}$**: 4-path none ($q$ has no 4-neighbour in $V$), 8-path 4, m-path 5. (b)** $V = \{1, 2\}$**: 4-path 6, 8-path 4, m-path 6.**

**Q3.** $p = (5, 1)$, $q = (1, 5)$, $V = \{1, 2, 3\}$.

**4-path: does not exist.** $p$'s 4-neighbours are $(4,1)=8$ and $(5,2)=4$, both outside $V$: $p$ cannot take a single 4-step.

**8-path (shortest, length 4):** straight up the anti-diagonal, $(5,1)\to(4,2)\to(3,3)\to(2,4)\to(1,5)$, values 2, 2, 1, 1, 2. (Six different 8-paths exist without repeating a pixel; this is the shortest.)

| | | | | |
|---|---|---|---|---|
| 5 | 8 | 3 | 4 | $2_q$ |
| 4 | 8 | 7 | 1 | 3 |
| 3 | 6 | 1 | 8 | 3 |
| 8 | 2 | 3 | 6 | 1 |
| $2_p$ | 4 | 6 | 9 | 3 |

**m-path (the only one, length 6).** Check each diagonal of the 8-path:

| diagonal | corners | m? |
|---|---|---|
| $(5,1)\to(4,2)$ | $(4,1)=8$, $(5,2)=4$ | allowed |
| $(4,2)\to(3,3)$ | $(3,2)=6$, $(4,3)=3\in V$ | refused → via $(4,3)$ |
| $(3,3)\to(2,4)$ | $(2,3)=7$, $(3,4)=8$ | allowed |
| $(2,4)\to(1,5)$ | $(1,4)=4$, $(2,5)=3\in V$ | refused → via $(2,5)$ |

Path $(5,1)\to(4,2)\to(4,3)\to(3,3)\to(2,4)\to(2,5)\to(1,5)$.

| | | | | |
|---|---|---|---|---|
| 5 | 8 | 3 | 4 | $2_q$ |
| 4 | 8 | 7 | 1 | 3 |
| 3 | 6 | 1 | 8 | 3 |
| 8 | 2 | 3 | 6 | 1 |
| $2_p$ | 4 | 6 | 9 | 3 |

**Q3: no 4-path ($p$ has no 4-neighbour in $V$); shortest 8-path length 4 along the anti-diagonal; m-path length 6 (unique), detouring through $(4,3)$ and $(2,5)$.**

↩ back to the card

### S3.2 · 2 · Shortest 4-, 8- and m-paths, and when none exists — Answers

(a) 6. (b) No: the 4 is in $V$, so the diagonal is refused. (c) No: every m-step is also an 8-step, so the shortest 8-path is never longer.

↩ back to the card

### S3.3 · 3 · Adjacent subsets and counting regions — Show the full exam answer

**Q1.** Only the border between column 5 (last column of $S_1$) and column 6 (first column of $S_2$) matters, rows 1–4.

1. 1s of $S_1$ in column 5: only $(4,5)$. 1s of $S_2$ in column 6: only $(3,6)$.
2. $(4,5)$ and $(3,6)$ touch only at a corner, and no other pair of 1s touches across the border, so the subsets are **not 4-adjacent**.
3. $(3,6)\in N_D\big((4,5)\big)$, so they are **8-adjacent**.
4. m: the shared 4-neighbours are $(3,5)=0$ and $(4,6)=0$, neither in $V$, so the diagonal is allowed: **m-adjacent**.

**Q1: (a) not 4-adjacent, (b) 8-adjacent, (c) m-adjacent.**

**Q4.** Group the 55 ones into connected components. Letters mark the regions (· = 0).

**4-adjacency: 16 regions.** Sizes A 4, B 1, C 1, D 4, E 1, F 1, G 4, H 16, I 1, J 1, K 2, L 1, M 1, N 4, O 11, P 2 (total 55). The ring of short strokes around the top-left stays in pieces because the strokes meet only at corners.

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| · | · | A | A | A | A | · | · | · | · | · | · |
| · | B | · | · | · | · | C | · | · | · | · | · |
| D | · | E | · | · | F | · | G | · | · | · | H |
| D | · | · | · | · | · | · | G | · | · | H | H |
| D | · | I | · | · | J | · | G | · | H | H | H |
| D | · | · | K | K | · | · | G | · | H | H | H |
| · | L | · | · | · | · | M | · | H | H | H | H |
| · | · | N | N | N | N | · | · | · | H | · | H |
| · | · | · | · | · | · | · | · | · | H | · | · |
| O | O | O | O | O | · | · | · | P | · | · | · |
| O | · | O | · | O | · | · | · | P | · | · | · |
| O | · | O | · | O | · | · | · | · | · | · | · |

**8-adjacency: 3 regions.** The corner contacts join the ring, the vertical stroke in column 8 and the big right-hand blob into one region of 40 pixels (A); the small arc inside the ring is region B (4 pixels: $(5,3), (5,6), (6,4), (6,5)$); the comb at the bottom-left is region C (11 pixels).

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| · | · | A | A | A | A | · | · | · | · | · | · |
| · | A | · | · | · | · | A | · | · | · | · | · |
| A | · | A | · | · | A | · | A | · | · | · | A |
| A | · | · | · | · | · | · | A | · | · | A | A |
| A | · | B | · | · | B | · | A | · | A | A | A |
| A | · | · | B | B | · | · | A | · | A | A | A |
| · | A | · | · | · | · | A | · | A | A | A | A |
| · | · | A | A | A | A | · | · | · | A | · | A |
| · | · | · | · | · | · | · | · | · | A | · | · |
| C | C | C | C | C | · | · | · | A | · | · | · |
| C | · | C | · | C | · | · | · | A | · | · | · |
| C | · | C | · | C | · | · | · | · | · | · | · |

**m-adjacency: the same 3 regions.** m refuses a diagonal only when a shared corner pixel is a 1, and then the two pixels are already connected through that corner, so the grouping cannot change.

**Q4: 4-adjacency 16 regions; 8-adjacency 3 regions (40, 4 and 11 pixels); m-adjacency 3 regions (same as 8).**

↩ back to the card

## S3.4 · 3 · Adjacent subsets and counting regions — Answers

(a) Different 4-regions; the same 8- and m-region. (b) No: 8 adds connections and never removes one, so regions can only merge.

↩ back to the card

### S3.5 · 5 · Quick recall — Answers

1. 8; 2 (the other two fall outside the image).
2. Allowed only if neither of the two shared corner pixels ($N_4(p) \cap N_4(q)$) has a value in $V$.
3. 8 (steps, not pixels).
4. Gaps 2 and 3: $D_4 = 5$, $D_8 = 3$, $D_e = \sqrt{13} \approx 3.61$.
5. m drops a diagonal only when a corner pixel in $V$ already links the two pixels, so connectivity is unchanged.

↩ back to the card

LECTURE 3 (PP. 20–42)

# Interpolation and operations on whole images

Tier A · must ⏱ 25 min Slides: Lec 3 pp. 20–42 · Sheet 4 Q1–Q3 ↗ Szeliski Ch 3

The first half of Lecture 3 was about single pixels and their neighbours. The second half does two jobs with whole images: **resizing** them (inventing values at positions that never existed) and **combining** them (adding, multiplying, masking, testing whether an operator is linear). Sheet 4 Q1–Q3 come straight from the second job, and each takes a few minutes once the idea is clear.

**What the exam can ask from this part**

- Zoom a small image by nearest neighbour (pixel replication) or bilinear interpolation; shrink by row/column deletion; bicubic = 16 neighbours.
- Array product vs matrix product of two images.
- Prove an operator is linear or not (sum: linear; max, gamma: not), with a counter-example.
- Image arithmetic: the 0–510 and −255–255 ranges, clipping; adding noise; ROI masks (Sheet 4 Q2–Q3).
- Set and logical operations on binary images; point vs local vs global operations.

⌛ **Only 20 minutes?** Do cards 3, 4 and 5 (the three sheet questions), then card 2.

📖 READ & UNDERSTAND · ⏱ 5 min · Tier A

## 1 · Zooming means inventing pixels: copy the nearest, or blend the four nearest

How do you make a 2×2 image into a 4×4 one when only four values exist? You estimate the rest. That is **interpolation**: using known pixel values to estimate values where there is no pixel yet (L3 p20). Zooming has two steps: create the new grid, then give each new pixel a grey level (L3 p21).

**Nearest neighbour** copies the closest original pixel, so a 2× zoom turns each pixel into a 2×2 block (**pixel replication**). The slide's example (L3 p24):

| | |
|---|---|
| 50 | 100 |
| 180 | 250 |

| | | | |
|---|---|---|---|
| 50 | 50 | 100 | 100 |
| 50 | 50 | 100 | 100 |
| 180 | 180 | 250 | 250 |
| 180 | 180 | 250 | 250 |

It is fast but blocky: the slides call this the **checkerboard effect** (L3 p22). **Shrinking** by a whole factor runs it backwards: delete rows and columns instead of copying them (L3 p25).

**Bilinear** blends the 4 nearest pixels with $v(x, y) = ax + by + cxy + d$ (L3 p26), $x$ = row, $y$ = column; the four neighbours give four equations for $a, b, c, d$. On the same image (made for this guide; the slides have no numbers), corners $(0,0) = 50$, $(0,1) = 100$, $(1,0) = 180$, $(1,1) = 250$:

1. $d = 50$; $b + d = 100 \Rightarrow b = 50$; $a + d = 180 \Rightarrow a = 130$; $a + b + c + d = 250 \Rightarrow c = 20$.
2. Centre $(0.5, 0.5)$: $130(0.5) + 50(0.5) + 20(0.25) + 50 = 145$, the average of the four corners.
3. Shortcut: interpolate along the top and bottom edges, then between them. Zoom to 3×3: $\begin{bmatrix} 50 & 75 & 100 \\ 115 & 145 & 175 \\ 180 & 215 & 250 \end{bmatrix}$.

4. New in-between values (75, 145, …) make it smoother than nearest neighbour.

**Bicubic** (L3 p28) uses the 16 nearest pixels and $v = \sum_{i=0}^{3}\sum_{j=0}^{3} a_{ij}x^i y^j$: 16 coefficients, smoothest and slowest, the photo-editor default.

🧭 **Like zooming into an old phone photo.** A cheap viewer makes each pixel a bigger square (nearest neighbour); a better one fades smoothly between neighbouring colours (bilinear, bicubic). *Where it breaks:* neither adds real detail; both only guess from what is already there. More: Wikipedia: Bilinear interpolation.

📖 READ & UNDERSTAND · ⏱ 6 min · Tier A

## 2 · Combining images: pixel by pixel, inside 0–255

"Multiply two images" means the **array product**: each pixel times the pixel at the same place (L3 p29), not the matrix product. $\begin{bmatrix}1&2\\3&4\end{bmatrix} \odot \begin{bmatrix}5&6\\7&8\end{bmatrix} = \begin{bmatrix}5&12\\21&32\end{bmatrix}$, whereas the matrix product is $\begin{bmatrix}19&22\\43&50\end{bmatrix}$.

An **operator** $H$ turns an image $f$ into $g = H[f]$. It is **linear** if $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for all same-size images and all constants (L3 p30). The sum of pixels is linear (L3 p31). The max is not (L3 p32): with $f_1 = \begin{bmatrix}0&2\\2&3\end{bmatrix}$, $f_2 = \begin{bmatrix}6&5\\4&7\end{bmatrix}$, $a_1 = 1$, $a_2 = -1$, the left side is $\max \begin{bmatrix}-6&-3\\-2&-4\end{bmatrix} = -2$ and the right is $3 - 7 = -4$. One counter-example settles it.

**Arithmetic** ($+, -, \times, \div$) works pixel by pixel on images of the same size (L3 p33). Two 8-bit images sum to 0…510 and differ by −255…255, so the slides **clip**: negatives → 0, above 255 → 255 (L3 p35). Example: $\begin{bmatrix}200&50\\30&120\end{bmatrix} - \begin{bmatrix}100&90\\30&10\end{bmatrix} = \begin{bmatrix}100&-40\\0&110\end{bmatrix} \to \begin{bmatrix}100&0\\0&110\end{bmatrix}$. (Gonzalez instead scales $f - \min$ to 0–255, giving $\begin{bmatrix}238&0\\68&255\end{bmatrix}$; it keeps −40 and 0 apart.) A **mask** of 1s inside a **region of interest (ROI)** and 0s elsewhere, multiplied in, keeps only the ROI (L3 p34).

A binary image can be read as the **set** of its 1-pixels, so set operations become logical ones: union $A \cup B$ = OR, intersection $A \cap B$ = AND, complement $A^c$ = NOT, difference $A - B = A \cap B^c$ = A AND NOT B (L3 p36–37). **Slide error (L3 p36):** "elements belonging to either A, B or both" is printed next to $D = A \cap B$; that sentence describes the **union**. The intersection is the elements in **both**.

Finally, classify an operation by how much input one output pixel sees (L3 p38): **point** (only its own pixel: negative, threshold, gamma), **local** (a neighbourhood: 3×3 averaging), **global** (the whole image: the image mean, the equalization mapping). The Lec 3 code (L3 p39–42) does all three kinds of step at once: threshold the grey rabbit at 0.5 (point), NOT it by swapping 0 and 1, then `img = BWimageMask * BWimage` (an array product with a mask).

🧭 **Like a stencil over a spray-painted wall.** Where the stencil is open (1) the paint shows, where it is solid (0) the wall stays bare: that is the ROI mask. *Where it breaks:* a real stencil is all-or-nothing; a mask could hold 0.5 and dim a region instead. More: Szeliski Ch 3.1.

✏️ PEN & PAPER · ⏱ 6 min · Tier A

## 3 · Is gamma a linear operator? (Sheet 4 Q1)

IN PLAIN WORDS

① The gamma transformation $s = c\,r^{\gamma}$ raises each pixel value $r$ to the power $\gamma$ and multiplies by $c$. Linear means $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for **every** pair of images and constants (card 2).

WARM-UP BY HAND

② $s = r^2$, one-pixel images $f_1 = f_2 = 1$, $a_1 = a_2 = 1$: mix then square, $(1+1)^2 = 4$; square then mix, $1^2 + 1^2 = 2$. Four is not two.

THE ONE IDEA

③ **One counter-example is a full proof of "not linear"**, because linearity must hold for every input.

BUILD IT

④ 1. Left side: $c\,(a_1 f_1 + a_2 f_2)^{\gamma}$.

Why: apply $H$ to the mixed image.

2. Right side: $a_1 c f_1^{\gamma} + a_2 c f_2^{\gamma}$.

Why: apply $H$ to each, then mix.

3. Pick numbers that make them differ.

Why: a power doesn't spread over a sum, $(x+y)^2 \neq x^2 + y^2$.

CHECK

⑤ With $\gamma = 1$ both sides are $c\,(a_1 f_1 + a_2 f_2)$: equal. So gamma is linear only when $\gamma = 1$; say so in the answer.

EXAM PROBLEM

**Your turn — Sheet 4 Q1.** The gamma transformation is $s = c\,r^{\gamma}$, with constants $c$ and $\gamma$. Is it a linear operator? Prove your answer.

✍ Show the full exam answer: S3b.1 (end of this lecture)

TRY ALONE

⑦ Is the negative $s = 255 - r$ linear?

✍ Answers: S3b.2 (end of this lecture)

🖍 PEN & PAPER · ⏱ 4 min · Tier A

## 4 · Making a noisy image (Sheet 4 Q2)

IN PLAIN WORDS

① A "corrupted" image is the clean image with noise added on top, pixel by pixel: $c(x,y) = b(x,y) + a(x,y)$, both images the same size (L3 p33).

WARM-UP BY HAND

② Clean $\begin{bmatrix} 200 & 0 \\ 0 & 60 \end{bmatrix}$ plus noise $\begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix} = \begin{bmatrix} 210 & 20 \\ 30 & 100 \end{bmatrix}$.

THE ONE IDEA

③ **Additive noise is an array sum**, kept inside 0–255.

BUILD IT

④ 1. Check the sizes match.

Why: pixel $(x, y)$ pairs with pixel $(x, y)$.

2. Add each pair. 3. Clip anything above 255 to 255 (L3 p35).

Why: an 8-bit pixel can't store 256 or more.

CHECK

⑤ The largest possible sum is $255 + 255 = 510$, so scan for values above 255 before you finish.

EXAM PROBLEM

**Your turn — Sheet 4 Q2.** Generate a corrupted image $c$ from the noisy image $a$ and the noiseless image $b$, and explain the process.

| a | | |
|---|---|---|
| 250 | 0 | 0 |
| 250 | 250 | 0 |
| 0 | 0 | 0 |

| b | | |
|---|---|---|
| 1 | 2 | 3 |
| 4 | 5 | 6 |
| 7 | 8 | 9 |

✍ Show the full exam answer: S3b.3 (end of this lecture)

TRY ALONE

⑦ Add noise $\begin{bmatrix} 250 & 0 \\ 0 & 0 \end{bmatrix}$ to $\begin{bmatrix} 10 & 5 \\ 5 & 5 \end{bmatrix}$. What is the top-left pixel?

✍ Answers: S3b.4 (end of this lecture)

✏ PEN & PAPER · ⏱ 4 min · Tier A

## 5 · Keeping a region of interest (Sheet 4 Q3)

IN PLAIN WORDS

① A **mask** is an image of 1s and 0s the same size as yours. The array product keeps every pixel under a 1 and blacks out every pixel under a 0; what survives is the **ROI** (L3 p34).

WARM-UP BY HAND

② $\begin{bmatrix} 9 & 8 & 7 \\ 6 & 5 & 4 \\ 3 & 2 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 9 & 8 & 0 \\ 6 & 5 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ : the top-left 2×2 survives.

THE ONE IDEA

③ **Times 1 keeps, times 0 erases**, using the array product, never the matrix product.

BUILD IT

④ 1. Line the mask up on the image. 2. Multiply each pair.

Why: the mask's 1s mark the ROI (the dental X-ray on L3 p34).

CHECK

⑤ Count the 1s in the mask: that is the most non-zero pixels the result can have.

EXAM PROBLEM

**Your turn — Sheet 4 Q3.** Generate the ROI $c$ from image $a$ and mask $b$; describe the method.

| a | | | |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
| 5 | 6 | 7 | 8 |
| 9 | 10 | 11 | 12 |
| 13 | 14 | 15 | 16 |

| b | | | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

✍ Show the full exam answer: S3b.5 (end of this lecture)

TRY ALONE

⑦ Which mask would keep only the four corners of $a$?

✍ Answers: S3b.6 (end of this lecture)

### ⚠ Traps

- "Not linear" needs one counter-example; "linear" needs the general algebra. Don't prove linearity with a single example.
- Multiplying two images (noise, masks) means the array product, never the matrix product.
- Arithmetic results outside 0–255 must be clipped (the slides) or scaled (Gonzalez) before you call the result an 8-bit image.
- L3 p36 labels "in A, B or both" as $A \cap B$: that is the union $A \cup B$; the intersection is "in both".
- L3 p21 says "two techniques" for zooming, then lists three (nearest, bilinear, bicubic).
- Bilinear's $x$ is the row and $y$ the column here, so $(1, 0)$ is the bottom-left pixel (180), not the top-right.

## Solutions and answers — Lecture 3 (pp. 20–42)

### S3b.1 · 3 · Is gamma a linear operator? (Sheet 4 Q1) — Show the full exam answer

Let $H[f] = cf^{\gamma}$ (applied to every pixel). $H$ is linear iff $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$ for all images and constants.

Left side: $c\,(a_1 f_1 + a_2 f_2)^{\gamma}$. Right side: $a_1 c f_1^{\gamma} + a_2 c f_2^{\gamma}$.

Counter-example: $c = 1$, $\gamma = 2$, $a_1 = a_2 = 1$, $f_1 = 2$, $f_2 = 3$. Left: $(2+3)^2 = 25$. Right: $2^2 + 3^2 = 13$. Since $25 \neq 13$, the condition fails.

(Homogeneity fails too: $H[3 \cdot 2] = 36$ but $3\,H[2] = 12$.)

**The gamma transformation is not linear (for $\gamma \neq 1$); it is linear only in the special case $\gamma = 1$, $s = c\,r$.**

↩ back to the card

### S3b.2 · 3 · Is gamma a linear operator? (Sheet 4 Q1) — Answers

No: with $f_1 = f_2 = 0$, $a_1 = a_2 = 1$, left $= 255$, right $= 510$. The constant 255 breaks it.

↩ back to the card

### S3b.3 · 4 · Making a noisy image (Sheet 4 Q2) — Show the full exam answer

Model of a corrupted image: $c(x,y) = b(x,y) + a(x,y)$, the noiseless image plus the noise, computed element by element (both are 3×3). Results above 255 would be clipped to 255.

Row 1: $1 + 250 = 251$, $2 + 0 = 2$, $3 + 0 = 3$. Row 2: $4 + 250 = 254$, $5 + 250 = 255$, $6 + 0 = 6$. Row 3: $7, 8, 9$ unchanged. No value exceeds 255, so no clipping is needed.

$$c = \begin{bmatrix} 251 & 2 & 3 \\ 254 & 255 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

↩ back to the card

### S3b.4 · 4 · Making a noisy image (Sheet 4 Q2) — Answers

$10 + 250 = 260 > 255$, so it is clipped to **255**.

↩ back to the card

### S3b.5 · 5 · Keeping a region of interest (Sheet 4 Q3) — Show the full exam answer

Method: element-wise (array) multiplication $c(x,y) = a(x,y) \cdot b(x,y)$. The mask $b$ is 1 inside the ROI (the central 2×2) and 0 elsewhere, so the ROI pixels are kept and every other pixel becomes 0.

$$c = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 6 & 7 & 0 \\ 0 & 10 & 11 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

↩ back to the card

### S3b.6 · 5 · Keeping a region of interest (Sheet 4 Q3) — Answers

1s at positions (1,1), (1,4), (4,1), (4,4), 0 elsewhere; the result keeps 1, 4, 13, 16.

↩ back to the card

LECTURE 4

# Image enhancement: point transforms and histograms

Tier A · must ⏱ 35 min Slides: Lec 4-Image Enhancement.pdf (38 p) · Sheet 4 Q4–Q6 are in Lecture 5 ↗ Szeliski Ch 3

Enhancement means making a picture more useful: pulling detail out of the shadows, boosting contrast, or making it easier on the eye (L4 p3). This lecture does it the simplest way, one pixel at a time: each grey level $r$ goes through a fixed curve and comes out as $s$. The histogram at the end is the bridge to Lecture 5, where the curve is read off the image itself.

**What the exam can ask from this lecture**

- Write and apply the negative, log and gamma transforms; say what each does to dark and bright values (L4 pp. 8–16).
- Min-max contrast stretching of a small image to 0–255, and the piecewise function's special cases (L4 pp. 18–21).
- Intensity-level slicing, both versions, on a small image (L4 p25).
- Histogram equalization by table on the 4 × 8 slide image (L4 pp. 31–32).

⏳ **Only 20 minutes?** Do cards 2 and 4, in that order.

📖 READ & UNDERSTAND · ⏱ 7 min · Tier A

## 1 · Every pixel through the same curve

The slides frame enhancement as $g(x, y) = T[f(x, y)]$ (L4 p4): $f$ is the input grey value at position $(x, y)$, $g$ the output, and $T$ an **operator** that looks at a **neighbourhood** (a small patch of pixels around $(x, y)$). Shrink the neighbourhood to the pixel alone and $T$ becomes a curve $s = T(r)$ from input level $r$ to output level $s$: a **point operation**. (The slides also mention working in a **transform domain**, e.g. Fourier: transform, edit, transform back, L4 pp. 5–6.)

🧭 **Like** a phone photo's brightness slider: every pixel is looked up in the same table, so two pixels with the same grey always end up the same. *Where it breaks:* blur and sharpen need the neighbours too, so they are not point operations (Lecture 6).

**Negative** (p9): $s = L - 1 - r$, where $L$ is the number of grey levels (256 for 8-bit). In 8-bit, $40 \to 215$; in a 4-bit image ($L = 16$), $3 \to 12$. It helps **white or grey detail buried in large black regions**.

**Log** (p11): $s = c \log(1 + r)$ spreads a **narrow band of dark inputs over a wide band of outputs** and squeezes the brights. With $c = 255 / \ln 256 = 45.99$, input $10 \to 45.99 \ln 11 = 110.27 \to 110$. Used on Fourier spectra, whose values span millions (p12).

**Gamma** (p14): $s = c\, r^{\gamma}$ on intensities scaled to 0–1. $\gamma < 1$ **brightens the darks**, $\gamma > 1$ darkens. With $\gamma = 0.4$, $64 \to 255\,(64/255)^{0.4} = 146.69 \to 147$; with $\gamma = 2.5$, $64 \to 8$.

**Slide code (L4 pp. 13, 29).** `np.log(1 + img)` on a uint8 image wraps 255 + 1 to 0, and $\log 0 = -\infty$; convert first with `np.log1p(img.astype(float))`. The code also reads `my_image` but transforms an undefined `img_gray`.

↗ Guide Tutorial 7, steps 1–3 ↗ Szeliski §3.1

✏️ PEN & PAPER · ⏱ 9 min · Tier A

## 2 · Min-max stretching: the dullest pixel to 0, the brightest to 255

IN PLAIN WORDS

① A low-contrast image uses only part of the grey scale, say 60 to 150. **Contrast stretching** (the slides also call it **normalization**, p19) maps that part onto the whole range with a straight line (L4 p21): $I_{new} = (I - Min)\,\dfrac{NewMax - NewMin}{Max - Min} + NewMin$. $I$ is the input pixel, $Min/Max$ the smallest and largest input values, $NewMin/NewMax$ the target range (0 and 255). Round to the nearest integer, halves up.

WARM-UP BY HAND

② Image $\begin{bmatrix}10 & 30\\ 20 & 50\end{bmatrix}$: $Min = 10$, $Max = 50$, factor $255/40 = 6.375$. Then $10 \to 0$, $20 \to 63.75 \to 64$, $30 \to 127.5 \to 128$, $50 \to 255$. Output $\begin{bmatrix}0 & 128\\ 64 & 255\end{bmatrix}$.

THE ONE IDEA

③ **Subtract $Min$ first, then multiply by one fixed factor.** The order of pixels never changes; only the gaps between them grow. Lecture 4's general version (p18) is a three-piece line through $(r_1, s_1)$ and $(r_2, s_2)$ with $r_1 \le r_2$, $s_1 \le s_2$: if $r_1 = s_1$ and $r_2 = s_2$ it is the identity (no change); if $r_1 = r_2$, $s_1 = 0$, $s_2 = L - 1$ it is **thresholding**, which outputs a binary image.

BUILD IT

④ 1. Find $Min$, $Max$.

Why: they pin the line's two ends.

2. Factor $= 255/(Max - Min)$.

Why: $Max - Min$ input steps must cover 255 output steps.

3. $(I - Min) \times$ factor for each pixel.

Why: the darkest pixel must land on 0.

4. Round, halves up.

Why: the slide rounds 124.67 to 125, not down to 124.

CHECK

⑤ $Min$ must give 0 and $Max$ must give 255. A value outside 0–255 means you subtracted the wrong end.

EXAM PROBLEM

**Your turn — Lec 4 p21.** An 8-bit image has range [60, 150]. Stretch it to [0, 255]:

| | | |
|---|---|---|
| 100 | 120 | 125 |
| 104 | 110 | 150 |
| 60 | 130 | 140 |

✍ Show the full exam answer: S4.1 (end of this lecture)

TRY ALONE

⑦ (a) Stretch 130 from [60, 150] to [0, 100]. (b) What does stretching do to an image that already has a 0 and a 255?

✍ Answers: S4.2 (end of this lecture)

**Slide wording (L4 p7).** It describes the S-shaped curve (darks pushed darker, brights lighter: contrast stretching) and then calls it a thresholding function. The S-curve is contrast stretching; the step function beside it is thresholding.

↗ Guide Tutorial 7, step 4 ↗ Szeliski §3.1

🖍 PEN & PAPER · ⏱ 6 min · Tier A

## 3 · Slicing: light up one band of greys

IN PLAIN WORDS

① **Intensity-level slicing** highlights every pixel whose level lies in a chosen band $[A, B]$ by setting it to 255. Version (a) **preserves the background**: pixels outside the band keep their value. Version (b) is **binary**: pixels outside become 0.

WARM-UP BY HAND

② Band 20–40 on $\begin{bmatrix} 10 & 25 \\ 40 & 60 \end{bmatrix}$: (a) $\begin{bmatrix} 10 & 255 \\ 255 & 60 \end{bmatrix}$, (b) $\begin{bmatrix} 0 & 255 \\ 255 & 0 \end{bmatrix}$.

THE ONE IDEA

③ **Each pixel asks one yes/no question: am I inside the band?** Yes → 255 in both versions; no → keep (a) or 0 (b). Decide the boundary values (is $B$ itself inside?) before you start, and write it down.

BUILD IT

④ 1. Mark every pixel inside the band.

Why: those are the only ones that become 255.

2. (a) copy the rest unchanged; (b) set the rest to 0.

Why: that is the only difference between the versions.

CHECK

⑤ Version (b) contains only 0 and 255, and its 255s sit exactly where version (a)'s do.

EXAM PROBLEM

**Your turn — Lec 4 p25.** Apply intensity-level slicing: (a) intensities between 50 and 100 → 255, preserve the background; (b) between 50 and 100 → 255, everything else → 0.

| | | | |
|---|---|---|---|
| 110 | 120 | 130 | 135 |
| 100 | 94 | 98 | 200 |
| 30 | 60 | 70 | 30 |
| 28 | 29 | 25 | 27 |

✍ Show the full exam answer: S4.3 (end of this lecture)

**Slide boundary (L4 p25).** The question says "between 50 and 100", yet both printed answers treat the pixel 100 as outside (it stays 100 in (a), becomes 0 in (b)). That only fits $50 \le r < 100$. In the exam, state which reading you used.

**TRY ALONE**

⑦ (a) Version (b), band 50 ≤ r < 100: what happens to 49? (b) Slides pp. 23–24 are titled "Contrast stretching: Example". What do they really show?

✍ Answers: S4.4 (end of this lecture)

↗ Guide Tutorial 7, step 5 (slicing)

🖍 PEN & PAPER · ⏱ 9 min · Tier A

## 4 · Histogram equalization by table

**IN PLAIN WORDS**

① The **histogram** counts pixels per grey level: $h(r_k) = n_k$, with $r_k$ the $k$-th level $(0 \dots L-1)$ and $n_k$ its count (L4 p26). Divided by the pixel count $MN$ it becomes the **normalized histogram** $p_r(r_k) = n_k/MN$, which sums to 1. Its shape tells the story (p27): piled left = dark image, piled right = light, narrow in the middle = low contrast, spread wide = high contrast. **Equalization** (p31) maps each level to $s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$, rounded, so the levels spread over the whole range.

**WARM-UP BY HAND**

② 2-bit image $(L=4)$ $\begin{bmatrix}0 & 0 & 1 & 1\\ 0 & 0 & 0 & 3\end{bmatrix}$, $MN = 8$, $L-1 = 3$.

| $r_k$ | $n_k$ | running $\sum n$ | $3 \times \sum n/8$ | $s_k$ |
|---|---|---|---|---|
| 0 | 5 | 5 | 1.875 | 2 |
| 1 | 2 | 7 | 2.625 | 3 |
| 2 | 0 | 7 | 2.625 | 3 |
| 3 | 1 | 8 | 3 | 3 |

New image $\begin{bmatrix}2 & 2 & 3 & 3\\ 2 & 2 & 2 & 3\end{bmatrix}$.

**THE ONE IDEA**

③ **A level's new position is the fraction of pixels at or below it, times $L-1$.** A crowded level makes the running sum jump, so it is pushed away from its neighbours; the top level always lands on $L-1$.

🧭 **Like** ranking students by percentile instead of raw mark: a cluster of near-identical marks gets spread over many percentiles. *Where it breaks:* everyone with the same mark must share one percentile, so the result is spread out, never perfectly flat.

**BUILD IT**

④ 1. Count $n_k$ for every level, empty ones too.

Why: an empty level still repeats the running sum.

2. Running sum of $n_k$, divide by $MN$.

Why: that is $\sum p_r$, the cumulative fraction.

3. Multiply by $L-1$ and round, halves up.
Why: it rescales 0–1 to the level range.

4. Replace every pixel $r_k$ by $s_k$.
Why: it is still a point operation.

**CHECK**

⑤ The last occupied level gives exactly $L-1$, and $s_k$ never decreases as $k$ grows.

**EXAM PROBLEM**

**Your turn — Lec 4 p32.** Apply histogram equalization to the 3-bit image ($L=8$, $MN = 4 \times 8 = 32$) whose four rows all read

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 2 | 2 | 3 | 4 | 4 |

✍ Show the full exam answer: S4.5 (end of this lecture)

**Slide error (L4 p32, row $r=1$).** The slide writes 7(0.357) = 2 and prints rows 1 2 2 4 4 5 7 7. Its own running sum is 0.375, and 7 × 0.375 = 2.625 → **3**. Correct rows: 1 3 3 4 4 5 7 7.

**TRY ALONE**

⑦ (a) Gonzalez Example 3.5 (L4 pp. 33–34): 3-bit, $MN = 4096$, $n = 790, 1023, 850, 656, 329, 245, 122, 81$. Find $s_0$ and $s_1$. (b) What does `plt.hist(img)` draw for a 2-D image (L4 p28)?

✍ Answers: S4.6 (end of this lecture)

Matching (specification) and Sheet 4 Q4–Q6 continue in Lecture 5.

↗ Guide Tutorial 8 (equalization stepper) ↗ Wikipedia: Histogram equalization

🔁 **RECALL DRILL · ⏱ 4 min · Tier A**

## 5 · Quick recall

1. Negative of 200 in an 8-bit image? Of 5 in a 3-bit image?
2. An image is too dark overall. Which helps: $\gamma = 0.5$ or $\gamma = 3$?
3. Piecewise stretching with $r_1 = r_2$, $s_1 = 0$, $s_2 = L-1$ gives what?
4. A histogram with all its mass between 100 and 140 (of 0–255) means what kind of image?
5. In an equalization table, what is $s_k$ for the highest occupied level, and why?

✍ Answers: S4.7 (end of this lecture)

### ⚠ Traps

- Using 255 in the negative of a non-8-bit image: it is $L-1$ (15 for 4-bit, 7 for 3-bit).
- Truncating instead of rounding in stretching: the slide gets 125 from 124.67; `np.uint8` gives 124.
- Copying the slide's equalization row 1 2 2 4 4 5 7 7: the correct row is 1 3 3 4 4 5 7 7.
- Leaving the slicing boundary unstated: the slide's "between 50 and 100" keeps 100 outside.
- Dividing by the number of levels instead of the pixel count $MN$ when normalizing a histogram.
- Running the slide's log code on uint8: $1 + 255$ wraps to 0. Convert to float first.

# Solutions and answers — Lecture 4

### S4.1 · 2 · Min-max stretching: the dullest pixel to 0, the brightest to 255 — Show the full exam answer

$Min = 60$, $Max = 150$, $NewMin = 0$, $NewMax = 255$; factor $= 255/90 = 2.8333$.

| $I$ | 60 | 100 | 104 | 110 | 120 | 125 | 130 | 140 | 150 |
|---|---|---|---|---|---|---|---|---|---|
| $I - 60$ | 0 | 40 | 44 | 50 | 60 | 65 | 70 | 80 | 90 |
| $\times 2.8333$ | 0 | 113.33 | 124.67 | 141.67 | 170 | 184.17 | 198.33 | 226.67 | 255 |
| rounded | 0 | 113 | 125 | 142 | 170 | 184 | 198 | 227 | 255 |

| | | |
|---|---|---|
| 113 | 170 | 184 |
| 125 | 142 | 255 |
| 0 | 198 | 227 |

$$I_{new} = \text{round}\big((I - 60) \cdot 255/90\big) \rightarrow \begin{bmatrix} 113 & 170 & 184 \\ 125 & 142 & 255 \\ 0 & 198 & 227 \end{bmatrix}$$ **(matches the slide)**

Note: `np.uint8(...)` truncates instead of rounding, giving 124, 141 and 226 for 104, 110 and 140. Round first if you code it.

↩ back to the card

### S4.2 · 2 · Min-max stretching: the dullest pixel to 0, the brightest to 255 — Answers

(a) $70 \times 100/90 = 77.78 \rightarrow 78$. (b) Nothing: the factor is 1 and $Min = 0$, so $I_{new} = I$. Lecture 5's equalization helps there.

↩ back to the card

### S4.3 · 3 · Slicing: light up one band of greys — Show the full exam answer

Inside the band (taking $50 \leq r < 100$, the slide's own reading): 94, 98, 60, 70. Everything else is outside, including 100.

(a) Preserve the background:

| | | | |
|---|---|---|---|
| 110 | 120 | 130 | 135 |
| 100 | 255 | 255 | 200 |
| 30 | 255 | 255 | 30 |
| 28 | 29 | 25 | 27 |

(b) Binary:

| | | | |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
| 0 | 255 | 255 | 0 |
| 0 | 255 | 255 | 0 |
| 0 | 0 | 0 | 0 |

**The four pixels 94, 98, 60, 70 → 255. (a) the rest unchanged; (b) the rest → 0. If "between 50 and 100" includes 100, the pixel at row 2, column 1 also becomes 255 in both.**

↩ back to the card

### S4.4 · 3 · Slicing: light up one band of greys — Answers

(a) Outside → 0. (b) Slicing: cameraman pixels between 100 and 150 set to 255, the rest set to 0 or kept.

↩ back to the card

### S4.5 · 4 · Histogram equalization by table — Show the full exam answer

| $r_k$ | $n_k$ | $p_r = n_k/32$ | $\sum p_r$ | $7 \times \sum p_r$ | $s_k$ |
|---|---|---|---|---|---|
| 0 | 4 | 0.125 | 0.125 | 0.875 | 1 |
| 1 | 8 | 0.25 | 0.375 | 2.625 | 3 |
| 2 | 8 | 0.25 | 0.625 | 4.375 | 4 |
| 3 | 4 | 0.125 | 0.75 | 5.25 | 5 |
| 4 | 8 | 0.25 | 1 | 7 | 7 |
| 5–7 | 0 | 0 | 1 | 7 | 7 |

Mapping 0 → 1, 1 → 3, 2 → 4, 3 → 5, 4 → 7. Every row becomes:

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 3 | 3 | 4 | 4 | 5 | 7 | 7 |

New histogram: $n(1) = 4$, $n(3) = 8$, $n(4) = 8$, $n(5) = 4$, $n(7) = 8$; the levels now span 1–7 instead of 0–4.

$s = 1, 3, 4, 5, 7$ **for** $r = 0 \ldots 4$**; each row →** **1 3 3 4 4 5** 7 7

↩ back to the card

### S4.6 · 4 · Histogram equalization by table — Answers

(a) $s_0 = 7 \times 790/4096 = 1.35 \to 1$; $s_1 = 7 \times 1813/4096 = 3.10 \to 3$. The full map is 1, 3, 5, 6, 6, 7, 7, 7 (the book prints 1.33, 3.08 because it rounds $p_r$ to two decimals first; same levels). (b) One histogram per column, 10 bins each, in many colours. Use `plt.hist(img.ravel(), bins=256)`.

↩ back to the card

### S4.7 · 5 · Quick recall — Answers

1. 55; $7 - 5 = 2$. 2. $\gamma = 0.5$: e.g. $64 \to 255\,(64/255)^{0.5} = 127.75 \to 128$. 3. Thresholding: a binary image. 4. Low contrast (narrow, central). 5. $L - 1$: the running sum has reached all pixels, $\sum p_r = 1$.

↩ back to the card

LECTURE 5

# Histogram processing: equalization and matching

Tier A · must ⏱ 45 min Slides: Lec 5-Histogram Processing.pdf (19 p) · Sheet 4 Q4–Q6, Sheet 5 Q1–Q2

↗ Wikipedia: Histogram matching

Lecture 4 changed each pixel with a formula you chose (negative, log, gamma, stretching). Lecture 5 lets the image choose: the formula is read off the image's own histogram. Equalization spreads the grey levels so the picture uses the whole range; matching bends them until the histogram looks like one you specify. Equalization was taught in Lecture 4 (the slide example on L5 p3 repeats Lec 4 p32, with the same 0.357 slip), so the cards go straight to the sheet problems.

**What the exam can ask from this lecture**

- Contrast-stretch a small image to 0–255 (Sheet 4 Q4).
- Equalize from a histogram or from an image: the full $p_r$, running-sum, $s_k$ table, the transformation, the new image and its histogram (Sheet 4 Q5–Q6, Lec 5 p3).
- Histogram matching: $s_k$, $G(z_q)$, the $s \to z$ lookup and the final $r \to z$ map (Sheet 5 Q1–Q2, Lec 5 pp. 10–15).
- Explain why equalization can wash out a dark image and how matching fixes it (Lec 5 pp. 16–18).

⌛ **Only 20 minutes?** Do cards 2 and 5, in that order.

PEN & PAPER · ⏱ 6 min · Tier A

## 1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4)

IN PLAIN WORDS

① **Histogram (contrast) stretching** is Lec 4's straight-line map: $s = (r - r_{\min}) \cdot \dfrac{255}{r_{\max} - r_{\min}}$, rounded with halves up, where $r_{\min}$, $r_{\max}$ are the image's smallest and largest values. The sheet gives no bit depth, so take Lec 4's 8-bit target 0–255.

WARM-UP BY HAND

② $\begin{bmatrix} 2 & 5 \\ 4 & 8 \end{bmatrix}$: $r_{\min} = 2$, $r_{\max} = 8$, slope $255/6 = 42.5$.

$2 \to 0$, $4 \to 2 \times 42.5 = 85$, $5 \to 127.5 \to 128$, $8 \to 255$, so the output is $\begin{bmatrix} 0 & 128 \\ 85 & 255 \end{bmatrix}$.

THE ONE IDEA

③ **Stretching is linear: it moves the histogram's ends to 0 and 255 and keeps its shape.** Equalization reshapes it.

BUILD IT

④ 1. Find $r_{\min}$, $r_{\max}$.

Why: they pin the line's ends.

2. Slope $= 255/(r_{\max} - r_{\min})$.

Why: the input span must cover 255 output steps.

3. $s = (r - r_{\min}) \times$ slope, rounded halves up.

Why: subtract first so the darkest pixel lands on 0; Lec 4 p21 rounds this way.

**CHECK**

⑤ The minimum comes out 0, the maximum 255, and the pixel order never changes.

**EXAM PROBLEM**

**Your turn — Sheet 4 Q4.** Apply histogram stretching to the image

| | | |
|---|---|---|
| 7 | 12 | 8 |
| 20 | 9 | 6 |
| 10 | 15 | 1 |

✍ Show the full exam answer: S5.1 (end of this lecture)

**TRY ALONE**

⑦ Stretch 35 from [10, 60] to [0, 255].

✍ Answers: S5.2 (end of this lecture)

↗ Guide Tutorial 7, step 4 (contrast stretching, worked 3 × 3)

🖍 PEN & PAPER · ⏱ 7 min · Tier A

## 2 · Equalizing when you are handed only the counts (Sheet 4 Q5)

**IN PLAIN WORDS**

① The exam may give only the histogram: $n_k$ pixels at level $r_k$, $MN$ pixels in all, $L$ levels. Equalization (Lec 4, L5 p2) needs nothing else: $s_k = (L-1)\sum_{j\le k} p_r(r_j)$, rounded halves up. Sheet 4 Q5 prints a histogram and no question; the natural task, solved here, is "equalize it".

**WARM-UP BY HAND**

② 2-bit ($L-1=3$), $n = 2, 1, 3, 2$, $MN = 8$. Running counts 2, 3, 6, 8.
$\times 3/8$: 0.75, 1.125, 2.25, 3 $\rightarrow s = 1, 1, 2, 3$; new histogram $n_1 = 3$, $n_2 = 3$, $n_3 = 2$.

**THE ONE IDEA**

③ **Work with running counts and divide once at the end:** $s_k = \frac{L-1}{MN}\sum_{j\le k} n_j$**.** It avoids rounding the $p_r$ column, and the last row must give exactly $L-1$.

**BUILD IT**

④ 1. $MN$ = sum of counts.

Why: divide by pixels, not by levels.

2. Running counts.

Why: $T(r_k)$ counts every pixel at or below level $k$.

3. Multiply by $(L-1)/MN$ and round.

Why: scale 0–1 to the level range, integers only.

4. Merge levels that land together.

Why: that gives the new histogram.

CHECK

⑤ The last $s$ is $L-1$, the column never decreases, the new counts still total $MN$.

EXAM PROBLEM

**Your turn — Sheet 4 Q5.** An image has 3 bits/pixel, so values run 0 to 7, with this histogram. Equalize it: give the transformation and the new histogram.

| Grey level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| Pixels | 10 | 8 | 9 | 2 | 14 | 1 | 5 | 2 |

✍ Show the full exam answer: S5.3 (end of this lecture)

TRY ALONE

⑦ Why is $s_3 = 4$ the same as $s_2$ in Q5?

✍ Answers: S5.4 (end of this lecture)

↗ Wikipedia: Histogram equalization (worked 8 × 8 example)

PEN & PAPER · ⏱ 7 min · Tier A

## 3 · From image to histogram to new image (Sheet 4 Q6)

IN PLAIN WORDS

① Now the input is the image itself. (a) Count each level and draw bars (level across, count up). (b) The **transformation** is the lookup table $r_k \to s_k$. (c) Rewrite the image through that lookup, pixel by pixel.

WARM-UP BY HAND

② 2-bit $\begin{bmatrix}0 & 0 & 3\\ 1 & 1 & 3\end{bmatrix}$: $n = 2, 2, 0, 2$, $MN = 6$, running 2, 4, 4, 6.
$\times 3/6 \to s = 1, 2, 2, 3$; new image $\begin{bmatrix}1 & 1 & 3\\ 2 & 2 & 3\end{bmatrix}$.

THE ONE IDEA

③ **Count carefully, and give empty levels a row too**: a level with $n_k = 0$ repeats the running sum above it. Most lost marks here are miscounts, not formula slips.

BUILD IT

④ 1. Tally each level, check the tally sums to $MN$.
Why: one missed pixel shifts every later running sum.
2. Fill the table, empty levels included.
Why: $T$ must say where every level goes.
3. Replace each pixel $r$ by $s_r$.
Why: equalization is a point operation; neighbours don't matter.

CHECK

⑤ The new image's counts are the old counts moved to new levels (warm-up: two 1s, two 2s, two 3s).

EXAM PROBLEM

**Your turn — Sheet 4 Q6.** A 3-bit 4 × 4 image:

| | | | |
|---|---|---|---|
| 0 | 0 | 0 | 4 |
| 1 | 1 | 1 | 5 |
| 1 | 2 | 2 | 7 |
| 2 | 2 | 2 | 7 |

(a) Plot the histogram of this image. (b) Define the transformation that will equalize the pixel histogram. (c) Find the values of the new 4 × 4 histogram-equalized image.

✍ Show the full exam answer: S5.5 (end of this lecture)

TRY ALONE

⑦ The Q6 image already spans 0–7. Would stretching change it? Does equalization?

✍ Answers: S5.6 (end of this lecture)

↗ Guide Tutorial 8, step 2 (the slide's example, done right)

📖 READ & UNDERSTAND · ⏱ 4 min · Tier A

## 4 · Matching: equalize both histograms, then meet in the middle

What if a flat histogram is the wrong goal? Lec 5 pp. 16–18 show Mars's moon Phobos: mostly black sky, a huge spike near 0. Equalization hands that spike a big share of the range and the picture comes out **washed out**. **Histogram matching** (specification) lets you choose the target shape $p_z(z)$ yourself. The method (L5 pp. 7–9) runs two equalizations and joins them; input levels are $r$, output levels are $z$.

1. Equalize the input: $s_k = T(r_k) = (L-1)\sum_{j \le k} p_r(r_j)$, rounded. 2. Equalize the **target** the same way: $G(z_q) = (L-1)\sum_{i \le q} p_z(z_i)$, rounded. 3. Both are now on the same "flat" scale, so set $G(z) = s$ and solve: $z = G^{-1}(s)$. In a table, that means: **for each $s_k$, find the $z_q$ whose $G(z_q)$ is closest to $s_k$**. 4. Chain them: $r_k \to s_k \to z_q$.

A tiny number: if $s_0 = 1$ and the $G$ row reads 0, 0, 0, 1, 2, 5, 6, 7, then $G(z_3) = 1$ matches exactly and every pixel at $r_0$ becomes 3. If $s = 3$, no $G$ equals 3; the nearest is $G(z_4) = 2$ (distance 1, against 2 for $G(z_5) = 5$), so $s = 3 \to z = 4$.

🧭 **Like** converting a grade between two schools through a percentile: "top 30% at my school" is looked up as "whatever mark is top 30% at yours". $s$ and $G$ are the percentiles. *Where it breaks:* marks are whole levels, so the exact percentile often doesn't exist and you take the nearest one, which is why the output histogram only approximates the target.

**Two rules the slides use without saying:** both $s_k$ and $G(z_q)$ are rounded to integers before comparing (p11: "$G(z_3) = 1.05 \to 1$"), and when several $z$ share the same $G$ value, take the smallest $z$ (Gonzalez's convention).

**Slide error (L5 pp. 11, 13–15).** The slides list $s_4 = 7$ and so map $r_4 \to 7$. Their own equalization slide (p4) has $s_4 = 6.23 \to 6$, and the correct running sum is $7 \times 3648/4096 = 6.23$. With $s_4 = 6$, $r_4 \to z_6$. Proof that 6 is right: only then do you get the slide's own "Actual $p_z$" column (p10): $z_6 = 0.16 + 0.08 = 0.24$ and $z_7 = 0.06 + 0.03 + 0.02 = 0.11$.

↗ Wikipedia: Histogram matching ↗ Guide Tutorial 8, step 5 (matching table)

PEN & PAPER · ⏱ 9 min · Tier A

## 5 · The full matching table, 3-bit (Sheet 5 Q1)

IN PLAIN WORDS

① Table A is the input histogram ($n_k$ pixels at level $r_k$); table B is the **reference** $p_z(z_q)$ you want the output to have. Give the map $r_k \to z_q$ and the histogram the output actually ends up with.

WARM-UP BY HAND

② 2-bit. Input $n = 2, 1, 4, 1$ ($MN = 8$) → 3× running/8 = 0.75, 1.125, 2.625, 3 → $s = 1, 1, 3, 3$.
Target $p_z = 0, 0.1, 0.3, 0.6$ → 3× running = 0, 0.3, 1.2, 3 → $G = 0, 0, 1, 3$.
$s = 1$ finds $G(z_2) = 1$, $s = 3$ finds $G(z_3) = 3$: $r = 0, 1, 2, 3 \to z = 2, 2, 3, 3$.

THE ONE IDEA

**③ Two equalization columns, then a lookup from $s$ into $G$; the answer is the $z$ where you land, never the $G$ value.**

BUILD IT

④ 1. $s_k$ from the input counts.
Why: puts the input on the flat scale.
2. $G(z_q)$ = running $p_z \times (L-1)$, rounded.
Why: puts the target on the same scale.
3. For each $s_k$, take the $z$ with the nearest $G$.
Why: $z = G^{-1}(s)$ by table lookup.
4. Add the counts landing on each $z$.
Why: that is the actual output histogram.

CHECK

⑤ The map never decreases; levels whose $G$ is 0 get no pixels unless some $s_k = 0$.

EXAM PROBLEM

**Your turn — Sheet 5 Q1.** Compute the histogram matching from the tables below; table A is the original image and table B is the reference (3-bit, $L = 8$).

| level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| A: count | 790 | 1023 | 850 | 656 | 329 | 245 | 122 | 81 |
| A: PDF | 0.19 | 0.25 | 0.21 | 0.16 | 0.08 | 0.06 | 0.03 | 0.02 |
| B: PDF | 0 | 0 | 0 | 0.15 | 0.2 | 0.3 | 0.2 | 0.15 |

✍ Show the full exam answer: S5.7 (end of this lecture)

TRY ALONE

⑦ In Q1, which output levels can never appear, and why?

✍ Answers: S5.8 (end of this lecture)

↗ Szeliski Ch 3.1 (histogram equalization)

✏️ PEN & PAPER · ⏱ 9 min · Tier A

## 6 · Sixteen levels and a tie every few rows (Sheet 5 Q2)

IN PLAIN WORDS

① Card 5's method on a 4-bit image ($L = 16$, multiplier $L - 1 = 15$). New: the **tie**, where $s$ sits exactly between two $G$ values (e.g. $s = 4$ between $G = 2$ and $G = 6$). The slides never meet one, so state your rule; here **ties go to the smaller** $z$, the direction of Gonzalez's "smallest $z$" rule.

WARM-UP BY HAND

② 2-bit, $n = 2, 2, 2, 2 \to s = 1, 2, 2, 3$; $p_z = 0.1, 0.2, 0, 0.7 \to 3\times$ running 0.3, 0.9, 0.9, 3 $\to G = 0, 1, 1, 3$.
$s = 1$: $G = 1$ at $z_1$ and $z_2 \to z_1$. $s = 2$: $G = 1$ and $G = 3$ both 1 away $\to$ smaller, $z_1$. $s = 3 \to z_3$.
Map $r = 0, 1, 2, 3 \to z = 1, 1, 1, 3$ (ties upward would give $1, 3, 3, 3$).

THE ONE IDEA

③ **Write your tie rule in one line at the top of the answer, then apply it every time.** A marker can follow a stated rule, not a coin toss.

BUILD IT

④ 1. Running counts × $15/MN$, rounded.
Why: two-decimal PDFs put level 1 at 1.50 instead of 1.51, a hair from a different rounding.
2. Running $p_z$ × 15, rounded $\to G$.
Why: the slides round $G$ before matching.
3. Nearest $G$; same $G$ or equal distance $\to$ smaller $z$.
Why: one fixed rule makes the map well defined.
4. Assemble $r \to z$.
Why: that is the requested matching.

CHECK

⑤ The map never decreases and the top level goes to 15.

EXAM PROBLEM

**Your turn — Sheet 5 Q2.** Compute the histogram matching for a 4-bit image (table A original, table B reference).

| level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| A: count | 2596 | 4096 | 3926 | 6692 | 3596 | 3396 | 3796 | 3196 |
| B: PDF | 0 | 0 | 0 | 0.05 | 0.07 | 0 | 0 | 0.25 |
| level | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
| A: count | 4296 | 4616 | 4746 | 3896 | 3796 | 4596 | 4396 | 4996 |
| B: PDF | 0.17 | 0.15 | 0.07 | 0.09 | 0.03 | 0.02 | 0.04 | 0.06 |

✍️ Show the full exam answer: S5.9 (end of this lecture)

**TRY ALONE**

⑦ In Q2, which $z$ levels with nonzero $p_z$ still receive no pixels?

✍ Answers: S5.10 (end of this lecture)

↗ Wikipedia: Histogram matching (the algorithm)

🔁 **RECALL DRILL · ⏱ 3 min · Tier B**

## 7 · Ten-second answers

1. You equalize the Sheet 4 Q5 result a second time. What changes?
2. Why did equalization wash out the Phobos photo (L5 p17), and what fixed it?
3. In matching, why can $s$ and $G$ be compared at all?

✍ Answers: S5.11 (end of this lecture)

### ⚠ Traps

- Dividing by the number of levels instead of the pixel count $MN$.
- Skipping empty levels in the table: their row repeats the running sum, and leaving them out shifts every $s_k$ after them.
- Rounding halves to even (as Python's `round(2.5) = 2` does): the slides always round halves up, e.g. 127.5 → 128, 13.5 → 14.
- Copying the slides' 3-bit example (row $r = 1$ gives 3, not 2) or the matching slides' $s_4 = 7$ (it is 6, so $r_4 \to z_6$ ).
- Answering a matching question with the $G$ value instead of the $z$ index where that $G$ sits.
- Leaving the tie rule unstated in a matching answer; Sheet 5 Q2 has four ties.

## Solutions and answers — Lecture 5

### S5.1 · 1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4) — Show the full exam answer

Target range 0–255 (8-bit). $r_{\min} = 1$, $r_{\max} = 20$, so $s = (r-1) \cdot \dfrac{255}{19}$, slope $= 13.4211$.

| $r$ | 1 | 6 | 7 | 8 | 9 | 10 | 12 | 15 | 20 |
|---|---|---|---|---|---|---|---|---|---|
| $r-1$ | 0 | 5 | 6 | 7 | 8 | 9 | 11 | 14 | 19 |
| $\times 255/19$ | 0 | 67.11 | 80.53 | 93.95 | 107.37 | 120.79 | 147.63 | 187.89 | 255 |
| $s$ | 0 | 67 | 81 | 94 | 107 | 121 | 148 | 188 | 255 |

Stretched image:

| | | |
|---|---|---|
| 81 | 148 | 94 |
| 255 | 107 | 67 |
| 121 | 188 | 0 |

$$s = \text{round}\big((r-1) \cdot 255/19\big) \rightarrow \begin{bmatrix} 81 & 148 & 94 \\ 255 & 107 & 67 \\ 121 & 188 & 0 \end{bmatrix}$$

↩ back to the card

### S5.2 · 1 · Stretching: pull the darkest pixel to 0 and the brightest to 255 (Sheet 4 Q4) — Answers

$25 \times 255/50 = 127.5 \rightarrow 128$.

↩ back to the card

### S5.3 · 2 · Equalizing when you are handed only the counts (Sheet 4 Q5) — Show the full exam answer

$L = 8$, $MN = 10 + 8 + 9 + 2 + 14 + 1 + 5 + 2 = 51$. $s_k = \dfrac{7}{51} \sum_{j=0}^{k} n_j$.

| $r_k$ | $n_k$ | $p_r(r_k)$ | CDF | $7\times$ CDF | $s_k$ |
|---|---|---|---|---|---|
| 0 | 10 | 0.196 | 10/51 = 0.196 | 1.37 | 1 |
| 1 | 8 | 0.157 | 18/51 = 0.353 | 2.47 | 2 |
| 2 | 9 | 0.176 | 27/51 = 0.529 | 3.71 | 4 |
| 3 | 2 | 0.039 | 29/51 = 0.569 | 3.98 | 4 |
| 4 | 14 | 0.275 | 43/51 = 0.843 | 5.90 | 6 |
| 5 | 1 | 0.020 | 44/51 = 0.863 | 6.04 | 6 |
| 6 | 5 | 0.098 | 49/51 = 0.961 | 6.73 | 7 |
| 7 | 2 | 0.039 | 51/51 = 1 | 7.00 | 7 |

Transformation: $0 \rightarrow 1, 1 \rightarrow 2, 2 \rightarrow 4, 3 \rightarrow 4, 4 \rightarrow 6, 5 \rightarrow 6, 6 \rightarrow 7, 7 \rightarrow 7$.

New histogram (merged levels add): level 1: 10, level 2: 8, level 4: 9 + 2 = 11, level 6: 14 + 1 = 15, level 7: 5 + 2 = 7; levels 0, 3, 5 are empty. Total 51 ✓.

$s = 1, 2, 4, 4, 6, 6, 7, 7$ **for** $r = 0 \ldots 7$**; equalized histogram** $n_s = 0, 10, 8, 0, 11, 0, 15, 7$ **for** $s = 0 \ldots 7$**.**

↩ back to the card

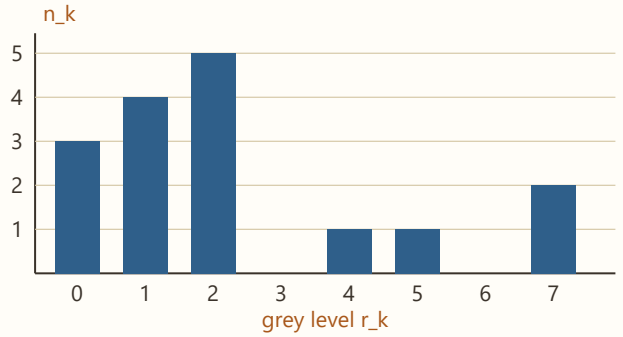
### S5.4 · 2 · Equalizing when you are handed only the counts (Sheet 4 Q5) — Answers

Level 3 holds only 2 of 51 pixels, so the running sum moves by $7 \times 2/51 = 0.27$ and stays in the same rounding bin.

↩ back to the card

### S5.5 · 3 · From image to histogram to new image (Sheet 4 Q6) — Show the full exam answer

(a) $L = 8$, $MN = 16$. Counts: $n = 3, 4, 5, 0, 1, 1, 0, 2$ for $r = 0 \ldots 7$ (sum 16 ✓).




(b) $s_k = \frac{7}{16} \sum_{j=0}^{k} n_j$, rounded:

| $r_k$ | $n_k$ | $p_r(r_k)$ | CDF | $7\times$ CDF | $s_k$ |
|---|---|---|---|---|---|
| 0 | 3 | 3/16 | 3/16 | 1.3125 | 1 |
| 1 | 4 | 4/16 | 7/16 | 3.0625 | 3 |
| 2 | 5 | 5/16 | 12/16 | 5.25 | 5 |
| 3 | 0 | 0 | 12/16 | 5.25 | 5 |
| 4 | 1 | 1/16 | 13/16 | 5.6875 | 6 |
| 5 | 1 | 1/16 | 14/16 | 6.125 | 6 |
| 6 | 0 | 0 | 14/16 | 6.125 | 6 |
| 7 | 2 | 2/16 | 16/16 | 7 | 7 |

Transformation: $0 \to 1, 1 \to 3, 2 \to 5, 3 \to 5, 4 \to 6, 5 \to 6, 6 \to 6, 7 \to 7$.

(c) Replace every pixel:

| | | | |
|---|---|---|---|
| 1 | 1 | 1 | 6 |
| 3 | 3 | 3 | 6 |
| 3 | 5 | 5 | 7 |
| 5 | 5 | 5 | 7 |

New histogram: level 1: 3, level 3: 4, level 5: 5, level 6: 2, level 7: 2 (total 16 ✓).

$T$: $(0, 1, 2, 3, 4, 5, 6, 7) \to (1, 3, 5, 5, 6, 6, 6, 7)$; **equalized image rows: 1 1 1 6 / 3 3 3 6 / 3 5 5 7 / 5 5 5 7.**

↩ back to the card

### S5.6 · 3 · From image to histogram to new image (Sheet 4 Q6) — Answers

Stretching: no ($r_{\min} = 0$, $r_{\max} = 7$, slope 1). Equalization: yes, the dark cluster 0–2 (12 of 16 pixels) spreads to 1, 3, 5 and the bright pixels squeeze into 6–7.

↩ back to the card

### S5.7 · 5 · The full matching table, 3-bit (Sheet 5 Q1) — Show the full exam answer

$MN = 4096$, $L - 1 = 7$. Rule: round $s_k$ and $G(z_q)$ to integers; map each $s_k$ to the $z_q$ with the closest $G$ (smallest $z$ if several share it).

**Step 1, equalize A:** $s_k = \frac{7}{4096} \sum_{j \le k} n_j$.

| $r_k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| running $n$ | 790 | 1813 | 2663 | 3319 | 3648 | 3893 | 4015 | 4096 |
| $7\times$ CDF | 1.35 | 3.10 | 4.55 | 5.67 | 6.23 | 6.65 | 6.86 | 7.00 |
| $s_k$ | 1 | 3 | 5 | 6 | 6 | 7 | 7 | 7 |

(Using the rounded PDF column instead gives 1.33, 3.08, 4.55, 5.67, 6.23, 6.65, 6.86, 7.00, the same $s_k$.)

**Step 2, equalize B:** $G(z_q) = 7 \sum_{i \le q} p_z(z_i)$.

| $z_q$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| CDF | 0 | 0 | 0 | 0.15 | 0.35 | 0.65 | 0.85 | 1 |
| $7\times$ CDF | 0 | 0 | 0 | 1.05 | 2.45 | 4.55 | 5.95 | 7.00 |
| $G(z_q)$ | 0 | 0 | 0 | 1 | 2 | 5 | 6 | 7 |

**Step 3,** $s \to z$**:** $s = 1 \to z_3$ ($G = 1$); $s = 3 \to z_4$ (no $G = 3$; nearest is $G = 2$ at distance 1, vs $G = 5$ at distance 2); $s = 5 \to z_5$; $s = 6 \to z_6$; $s = 7 \to z_7$.

**Step 4,** $r \to z$**:**

| $r_k$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| $z_q$ | 3 | 4 | 5 | 6 | 6 | 7 | 7 | 7 |

**Resulting histogram:** $z_3$: 790 (0.19), $z_4$: 1023 (0.25), $z_5$: 850 (0.21), $z_6$: 656 + 329 = 985 (0.24), $z_7$: 245 + 122 + 81 = 448 (0.11); $z_0$–$z_2$: 0. Close to the specified 0.15, 0.2, 0.3, 0.2, 0.15 but not equal: whole input levels can't be split.

> $r = 0 \ldots 7 \to z = 3, 4, 5, 6, 6, 7, 7, 7$**; output PDF** $p_z = 0, 0, 0, 0.19, 0.25, 0.21, 0.24, 0.11$**. (The slides'** $4 \to 7$ **is a slip:** $s_4 = 6.23 \to 6$**.)**

↩ back to the card

### S5.8 · 5 · The full matching table, 3-bit (Sheet 5 Q1) — Answers

$z_0, z_1, z_2$: their $G$ is 0, and no $s_k$ is 0 (the smallest is 1), so they never win the lookup.

↩ back to the card

### S5.9 · 6 · Sixteen levels and a tie every few rows (Sheet 5 Q2) — Show the full exam answer

$MN = 66\,632$, $L - 1 = 15$. Rule: round $s$ and $G$ to integers (halves up); map each $s$ to the $z$ with the nearest $G$; if several $z$ qualify (same $G$ or equal distance), take the smallest $z$.

**Step 1, equalize A:** $s_k = \frac{15}{66\,632} \sum_{j \le k} n_j$.

| $r_k$ | running $n$ | $15\times$ CDF | $s_k$ | $z_q$ | $\text{CDF}_z$ | $15\times$ $\text{CDF}_z$ | $G(z_q)$ |
|---|---|---|---|---|---|---|---|
| 0 | 2596 | 0.58 | 1 | 0 | 0 | 0 | 0 |
| 1 | 6692 | 1.51 | 2 | 1 | 0 | 0 | 0 |
| 2 | 10618 | 2.39 | 2 | 2 | 0 | 0 | 0 |
| 3 | 17310 | 3.90 | 4 | 3 | 0.05 | 0.75 | 1 |
| 4 | 20906 | 4.71 | 5 | 4 | 0.12 | 1.80 | 2 |
| 5 | 24302 | 5.47 | 5 | 5 | 0.12 | 1.80 | 2 |
| 6 | 28098 | 6.33 | 6 | 6 | 0.12 | 1.80 | 2 |
| 7 | 31294 | 7.04 | 7 | 7 | 0.37 | 5.55 | 6 |
| 8 | 35590 | 8.01 | 8 | 8 | 0.54 | 8.10 | 8 |
| 9 | 40206 | 9.05 | 9 | 9 | 0.69 | 10.35 | 10 |
| 10 | 44952 | 10.12 | 10 | 10 | 0.76 | 11.40 | 11 |
| 11 | 48848 | 11.00 | 11 | 11 | 0.85 | 12.75 | 13 |
| 12 | 52644 | 11.85 | 12 | 12 | 0.88 | 13.20 | 13 |
| 13 | 57240 | 12.89 | 13 | 13 | 0.90 | 13.50 | 14 |
| 14 | 61636 | 13.88 | 14 | 14 | 0.94 | 14.10 | 14 |
| 15 | 66632 | 15.00 | 15 | 15 | 1 | 15 | 15 |

**Step 2,** $s \to z$ ($G$ row: 0 0 0 1 2 2 2 6 8 10 11 13 13 14 14 15):

| $s$ | nearest $G$ (at $z$) | $z$ |
|---|---|---|
| 1 | 1 ($z_3$) | 3 |
| 2 | 2 ($z_4, z_5, z_6$) → smallest | 4 |
| 4 | tie: 2 ($z_4$) and 6 ($z_7$), both 2 away → smaller | 4 |
| 5 | 6 ($z_7$), 1 away | 7 |
| 6 | 6 ($z_7$) | 7 |
| 7 | tie: 6 ($z_7$) and 8 ($z_8$) → smaller | 7 |
| 8 | 8 ($z_8$) | 8 |
| 9 | tie: 8 ($z_8$) and 10 ($z_9$) → smaller | 8 |
| 10 | 10 ($z_9$) | 9 |
| 11 | 11 ($z_{10}$) | 10 |
| 12 | tie: 11 ($z_{10}$) and 13 ($z_{11}$) → smaller | 10 |
| 13 | 13 ($z_{11}, z_{12}$) → smallest | 11 |
| 14 | 14 ($z_{13}, z_{14}$) → smallest | 13 |
| 15 | 15 ($z_{15}$) | 15 |

**Step 3,** $r \to z$**:**

| $r$ | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $z$ | 3 | 4 | 4 | 4 | 7 | 7 | 7 | 7 | 8 | 8 | 9 | 10 | 10 | 11 | 13 | 15 |

Output counts: $z_3$ 2596, $z_4$ 14714, $z_7$ 13984, $z_8$ 8912, $z_9$ 4746, $z_{10}$ 7692, $z_{11}$ 4596, $z_{13}$ 4396, $z_{15}$ 4996 (all others 0; total 66 632).

$r = 0 \ldots 15 \to z = 3, 4, 4, 4, 7, 7, 7, 7, 8, 8, 9, 10, 10, 11, 13, 15$ **(ties to the smaller $z$). If ties are**

**broken upward, only four entries change:** $r_3 \to 7$, $r_7 \to 8$, $r_9 \to 9$, $r_{12} \to 11$.

↩ back to the card

### S5.10 · 6 · Sixteen levels and a tie every few rows (Sheet 5 Q2) — Answers

$z_{12}$ and $z_{14}$: their $G$ values (13 and 14) are already claimed by the smaller $z_{11}$ and $z_{13}$.

↩ back to the card

### S5.11 · 7 · Ten-second answers — Answers

1. Nothing: every occupied level (1, 2, 4, 6, 7) maps to itself. 2. The huge dark-sky spike near 0 took a big share of the output range and pushed everything up to light grey; matching to a specified histogram fixed it, because its mapping (curve 2 on L5 p18) keeps dark input pixels dark. 3. Both are equalizations onto the same flat 0 … $L-1$ scale, $s = T(r)$ and $G(z)$, so equal values mean "same position in the distribution".

↩ back to the card

LECTURE 6

# Spatial filtering: correlation, convolution, smoothing

Tier A · must ⏱ 55 min Slides: Lec 6-Spatial Filtering.pdf (31 p) · Sheet 5 Q3–Q5, Sheet 6 Q1–Q6 ↗ Szeliski Ch 3

Lectures 4 and 5 changed each pixel on its own: a lookup table, a histogram. Now every new pixel is computed from a small **neighbourhood** of old pixels, which is how blur, noise removal and (next lecture) sharpening all work. Almost every exam number in this lecture is a sum of nine products or a sorted list of nine values, so the skill is careful bookkeeping.

**What the exam can ask from this lecture**

- Turn a correlation kernel into the convolution kernel that gives the same output (rotate 180°), in 1-D and 2-D.
- Apply a filter to a small sub-image: the output size, then every output value.
- Pad an image for a given filter size (how many rows/columns, what values).
- Blur with the 3×3 averaging filter, with and without zero padding.
- 3×3 median filter with zero padding; why the median beats the mean on salt-and-pepper noise.
- Definitions: kernel/mask, linear vs nonlinear (order-statistic) filter, box vs weighted average, low-pass.

⌛ **Only 20 minutes?** Do cards 2, 5 and 6, in that order.

📖 READ & UNDERSTAND · ⏱ 8 min · Tier A

## 1 · A filter is a small grid of weights that slides over the image

Picture a 3×3 window on the image. The nine pixels under it are the pixel's **neighbourhood**. A **spatial filter** (also **mask**, **kernel**, template, window, L6 p3) is that neighbourhood plus an operation; the result goes into the output at the window's **centre**, then the window moves one pixel.

"Multiply by weights and add" makes the filter **linear** (L6 p4):

$$g(x,y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s,t)\, f(x+s,\ y+t), \qquad a = \tfrac{m-1}{2},\ b = \tfrac{n-1}{2}$$

$f$ is the input, $g$ the output, $w$ the $m \times n$ kernel with $(0,0)$ at its centre. Kernels have **odd sizes** (smallest 3×3) so there is a centre to write into. Tiny example: patch 1 2 0 / 3 1 1 / 0 2 4 with kernel 1 0 1 / 0 2 0 / 1 0 1 gives $1 \cdot 1 + 2 \cdot 1 + 1 \cdot 4 = 7$ (only the non-zero weights count). Sorting the nine values and keeping one (min, max, **median**) is a **nonlinear**, order-statistic filter (L6 p7, p18).

A **low-pass** (smoothing) filter blurs: it reduces noise and small details (L6 p10–p11). Its weights are all positive and **sum to 1**, so flat regions keep their brightness. Two masks: the **box filter** $\frac{1}{9}\times$(all ones), where every neighbour counts equally, and the **weighted average** $\frac{1}{16}\times$ 1 2 1 / 2 4 2 / 1 2 1, which trusts the centre most and blurs less.

The slides' example (L6 p13–p15) is a 4×4 image, left half 10 and right half 200, with border padding. Column 2 sees six 10s and three 200s: $\frac{660}{9} = 73.3 \to$ **73**; column 3: $\frac{1230}{9} = 136.7 \to$ **137**. The step becomes a ramp 10, 73, 137, 200. The weighted mask gives $\frac{920}{16} = 57.5 \to 58$ and $\frac{2440}{16} = 152.5 \to 153$. The slides **round to the nearest integer**, halves up. The zero-padded "?" on L6 p16–p17 works out to box: 4, 49, 91, 89 (top/bottom rows) and 7, 73, 137, 133 (middle rows); weighted: 6, 43, 114, 113 and 8, 58, 153, 150. The borders go dark because the zeros are averaged in.

⚠️ **Slide error (L6 p15, p17).** These slides print the weighted mask with $\frac{1}{9}$. Its weights sum to 16, so the factor must be $\frac{1}{16}$, as on L6 p11 and p13. The slides' own 58 and 153 only come out with $\frac{1}{16}$.

🧭 **Like a magnifying glass sliding over a page.** At each stop you see one small patch and write one summary of it. *Where it breaks:* the glass only shows the patch, while a filter also weights each cell, and those weights decide whether it blurs or sharpens.

More: Wikipedia: Kernel (image processing), Gaussian blur (the 1-2-1 mask approximates one).

✏️ PEN & PAPER · ⏱ 9 min · Tier A

## 2 · Correlation and convolution differ by one 180° turn

IN PLAIN WORDS

① Card 1's formula is **correlation** ($w \star f$ with a hollow star): slide the kernel as printed. **Convolution** (solid star) first **rotates the kernel 180°** (flip left–right and up–down), shown by the minus signs in $\sum\sum w(s,t)\, f(x-s, y-t)$ (L6 p8). Sheet 6 Q1–Q2 give a correlation kernel and ask for the convolution kernel that gives the same output.

WARM-UP BY HAND

② Kernel [2 0 6] on an impulse [0 0 1 0 0] (zero padded). Correlation gives [0 6 0 2 0], the kernel backwards. Convolution gives [0 2 0 6 0], a true copy (2-D version: L6 p9).

THE ONE IDEA

**③ Convolution with $w$ = correlation with $w$ rotated 180°. Hand the convolution the rotated kernel and the two rotations cancel.**

BUILD IT

④ on the sibling $\begin{bmatrix} 2 & 0 & 1 \\ 3 & 5 & 4 \\ 9 & 6 & 8 \end{bmatrix}$.

1. Flip each row left–right: $\begin{bmatrix} 1 & 0 & 2 \\ 4 & 5 & 3 \\ 8 & 6 & 9 \end{bmatrix}$.
   Why: first half of the turn (in 1-D this is the whole turn: [2 0 6] → [6 0 2]).
2. Reverse the row order: $\begin{bmatrix} 8 & 6 & 9 \\ 4 & 5 & 3 \\ 1 & 0 & 2 \end{bmatrix}$.
   Why: two mirror flips make a 180° rotation, and the centre stays put.

CHECK

⑤ On $f = [3\ 1\ 4\ 1\ 5]$, correlation with [2 0 6] and convolution with [6 0 2] both give [30 8 38]. A kernel that a half-turn leaves unchanged (box, weighted) gives the same output either way. More: Wikipedia: Convolution.

🧭 **Like a rubber stamp.** The carved letters are mirrored, so pressing the stamp (convolution) prints them the right way round. *Where it breaks:* a stamp mirrors only left–right, while the kernel also flips

up–down.

EXAM PROBLEM

**Your turn — Sheet 6 Q1.** Given the correlation matrix below, convert it into a convolution matrix so the output is the same as the correlation, and specify the method you use.

| | | |
|---|---|---|
| 1 | 7 | 3 |

**Sheet 6 Q2.** Do the same for the 3×3 correlation matrix:

| | | |
|---|---|---|
| 1 | 7 | 3 |
| 8 | 8 | 7 |
| 1 | 4 | 5 |

✍ Show the full exam answer: S6.1 (end of this lecture)

TRY ALONE

⑦ Rotate the column $[4; 5; 6]$ by 180°.

✍ Answers: S6.2 (end of this lecture)

🖍 PEN & PAPER · ⏱ 10 min · Tier A

## 3 · Applying a filter: first the size, then the sums

IN PLAIN WORDS

① Sheet 6 Q3–Q4 give a **sub-image** and a filter and ask for the new size and the output. No padding is mentioned, so the window only sits where it fits completely ("ignore the edges", L6 p6) and the output **shrinks**. "Applying a filter" means the slides' formula (L6 p4), which is **correlation**.

WARM-UP BY HAND

② $[2\ 1\ 0\ 3]$ with filter $[1\ 0\ 2]$: the window fits twice. $2 \cdot 1 + 1 \cdot 0 + 0 \cdot 2 = 2$ and $1 \cdot 1 + 0 \cdot 0 + 3 \cdot 2 = 7$ give $[2\ 7]$: four inputs, two outputs.

THE ONE IDEA

③ **Output size $= n - k + 1$ per direction (input length $n$, kernel length $k$). With $p$ zeros on each side it becomes $n + 2p - k + 1$.** The slides only say "smaller"; the formula comes from counting where the window fits.

BUILD IT

④ on a 3×4 sibling $\begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 \end{bmatrix}$ with kernel $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$.

1. Size $(3 - 3 + 1) \times (4 - 3 + 1) = 1 \times 2$.
   Why: the window fits in one vertical position and two horizontal ones.
2. Columns 1–3, using only the non-zero weights: $1 \cdot 1 + 2 \cdot 2 + 1 \cdot 1 + 1 \cdot 1 = 7$.
   Why: zero weights wipe out their pixel, so skip them.
3. Columns 2–4: $1 \cdot 2 + 2 \cdot 0 + 1 \cdot 3 + 1 \cdot 1 = 6$, so the output is $[7\ 6]$.

Why: the same four kernel positions on new pixels.

**CHECK**

⑤ Two values for a 1×2 output. Convolution would give [3 8], a different answer, so name the operation you used. Szeliski Ch 3.2 (szeliski.org/Book) works the same sums.

🧭 **Like a stencil on a tiled floor.** A 3-tile stencil on a 5-tile row fits in only 3 places, so you get 3 readings. *Where it breaks:* padding lets the stencil hang over the edge, which a real floor can't.

**EXAM PROBLEM**

**Your turn — Sheet 6 Q3.** Given the sub-image and filter below, what is the new size of the sub-image after applying the filter? Display the output.

| | | | | |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 |

filter

| | | |
|---|---|---|
| 3 | 4 | 5 |

**Sheet 6 Q4.** Same question for

| | | | | |
|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 0 |

filter

| | | |
|---|---|---|
| 3 | 4 | 5 |
| 1 | 0 | 1 |
| 0 | 4 | 0 |

✍️ Show the full exam answer: S6.3 (end of this lecture)

**TRY ALONE**

⑦ What is the output size for a 6-long row and a 5-long filter, with 2 zeros on each side?

✍️ Answers: S6.4 (end of this lecture)

🖍 **PEN & PAPER · ⏱ 6 min · Tier A**

## 4 · Padding: give the window something to stand on at the border

**IN PLAIN WORDS**

① On a border pixel the window hangs outside the image (L6 p6). **Padding** adds a frame of extra pixels so that every pixel can be a centre and the output keeps the input's size. Sheet 6 Q5–Q6 ask for the padding for filters of size 3 and 3×3.

**WARM-UP BY HAND**

② [3 1 4] with a 3-long filter becomes [0 3 1 4 0]. With a 5-long filter it becomes [0 0 3 1 4 0 0].

THE ONE IDEA

**③ Pad $p = \frac{k-1}{2}$ on *every* side for a $k \times k$ filter. The output size $n + 2p - k + 1$ then equals $n$.**

BUILD IT

④ on $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ with a 3×3 filter.

1. $p = (3-1)/2 = 1$.
   Why: the window reaches one pixel past its centre.
2. Add a ring of zeros, going from 2×2 to 4×4: $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 3 & 4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.
   Why: $+2p$ per dimension; the original values stay untouched.

CHECK

⑤ A 3×3 filter over the 4×4 array gives $4 - 3 + 1 = 2$, so the output is back to 2×2. The slides also use **border padding**, which copies the edge pixel outward (L6 p14, p21): [3 1 4] → [3 3 1 4 4]. Zero padding darkens the borders, the "unwanted artifacts" of L6 p6. Other border rules are under "Edge handling" in Wikipedia: Kernel (image processing).

🧭 **Like the mount around a framed photo.** It adds a border without covering the picture. *Where it breaks:* a mount is never averaged into the photo, but zero padding is.

EXAM PROBLEM

**Your turn — Sheet 6 Q5.** Given the sub-image below, calculate its padding if the filter size is 3.

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 2 | 1 | 2 | 1 | 2 | 1 |

**Sheet 6 Q6.** Calculate the padding if the filter size is 3×3:

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 1 | 2 | 1 |
| 2 | 1 | 2 | 1 | 2 |
| 1 | 2 | 1 | 2 | 1 |
| 2 | 1 | 2 | 1 | 2 |
| 1 | 2 | 1 | 2 | 1 |

✍️ Show the full exam answer: S6.5 (end of this lecture)

TRY ALONE

⑦ A 10×10 image padded for a 5×5 filter: what size is the padded image?

✍️ Answers: S6.6 (end of this lecture)

🖍 PEN & PAPER · ⏱ 10 min · Tier A

## 5 · Box blur by hand, without padding and with zeros

IN PLAIN WORDS

① In Sheet 5 Q3 ("blur using a 3×3 filter") no padding is mentioned, so the output is the 3×3 interior (card 3). Q4 pads with zeros, so the output stays 5×5 (card 4). The filter is the slides' **box filter**: each output is the sum of the nine pixels under the window ÷ 9, rounded to the nearest integer.

**WARM-UP BY HAND**

② 4×4 sibling 1 3 2 0 / 4 2 1 3 / 0 1 5 2 / 2 2 0 1. The four window sums are 19, 19, 17, 17; ÷ 9 gives 2.11 and 1.89, all rounding to 2. With zero padding, the top-left corner holds only $1 + 3 + 4 + 2 = 10$, and $10/9 \to 1$.

**THE ONE IDEA**

③ **Add column sums, not nine separate numbers.** For each band of three rows, write each column's 3-pixel sum once. Each window sum is then three neighbouring column sums added together.

**BUILD IT**

④ on the sibling.

1. Rows 1–3 column sums: 5, 6, 8, 5.
   Why: each column is shared by up to three windows.
2. Window sums: $5 + 6 + 8 = 19$ and $6 + 8 + 5 = 19$, then $19/9 = 2.11 \to 2$.
   Why: the box mask weights every cell by $\frac{1}{9}$.
3. Rows 2–4: column sums 6, 5, 6, 6 give windows 17, 17, and $17/9 = 1.89 \to 2$.
   Why: slide the band down one row and repeat.

**CHECK**

⑤ Each blurred value lies between its window's min and max. With zero padding the border values come out lower than the interior ones. The box filter is the first entry in Wikipedia's kernel table.

🧭 **Like blurry CCTV.** Each dot on screen is the average of a patch of the scene, so a face becomes a smooth blob. *Where it breaks:* CCTV loses detail because it has few dots; here the pixel count stays the same and the detail is averaged away.

**EXAM PROBLEM**

**Your turn — Sheet 5 Q3.** Given the sub-image below, blur it using a 3×3 filter.

| | | | | |
|---|---|---|---|---|
| 5 | 2 | 4 | 2 | 1 |
| 2 | 1 | 2 | 4 | 2 |
| 5 | 4 | 5 | 2 | 1 |
| 2 | 1 | 4 | 5 | 2 |
| 1 | 2 | 1 | 2 | 5 |

**Sheet 5 Q4.** Blur the same sub-image with padding of 0s using a 3×3 filter.

✍ Show the full exam answer: S6.7 (end of this lecture)

**TRY ALONE**

⑦ In Q4, which output is the smallest, and why there?

✍ Answers: S6.8 (end of this lecture)

PEN & PAPER · ⏱ 8 min · Tier A

## 6 · The median filter: sort nine values, keep the middle one

IN PLAIN WORDS

① Sheet 5 Q5 asks for a 3×3 **median filter** with zero padding. For each pixel, list the nine values under the window (padding zeros included), sort them, and write down the 5th one. There are no weights and no division: it is an order-statistic (nonlinear) filter (L6 p18).

WARM-UP BY HAND

② Take the window 10 12 11 / 13 **255** 12 / 11 10 12, where 255 is a "salt" pixel. The mean is $346/9 \rightarrow 38$, a grey smudge. The median is 10 10 11 11 **12** 12 12 13 255 → 12, a value that was really there.

THE ONE IDEA

③ **Sort the nine values and take position 5.** One extreme pixel only ever moves to an end of the list, so the median removes **salt-and-pepper** noise while keeping edges (L6 p18–p20).

BUILD IT

④ on $\begin{bmatrix} 1 & 9 & 2 \\ 3 & 0 & 4 \\ 5 & 6 & 7 \end{bmatrix}$ with zero padding.

1. Centre: 0 1 2 3 **4** 5 6 7 9 → 4.
   Why: the window is fully inside the image.
2. Top-left: 1, 9, 3, 0 plus five zeros: 0 0 0 0 **0** 0 1 3 9 → 0.
   Why: five padding zeros fill sorted positions 1–5.
3. Top-middle: 0 0 0 0 **1** 2 3 4 9 → 1. All nine: $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 4 & 2 \\ 0 & 3 & 0 \end{bmatrix}$.
   Why: an edge window has three padding zeros.

CHECK

⑤ Every output is one of its nine window values, and with zero padding all four corners come out 0. The slides' own example (L6 p21–p23) uses border padding, which avoids those dark corners. More: Wikipedia: Median filter.

🧭 **Like dropping the highest and lowest judges' scores.** One crazy score can't move the result. *Where it breaks:* the median keeps only the single middle score instead of averaging the rest.

EXAM PROBLEM

**Your turn — Sheet 5 Q5.** Given the sub-image below, blur it with padding of 0s using a 3×3 median filter.

| | | | | |
|---|---|---|---|---|
| 5 | 2 | 4 | 2 | 1 |
| 2 | 1 | 2 | 4 | 2 |
| 5 | 4 | 5 | 2 | 1 |
| 2 | 1 | 4 | 5 | 2 |
| 1 | 2 | 1 | 2 | 5 |

✍ Show the full exam answer: S6.9 (end of this lecture)

**TRY ALONE**

⑦ What is the median of 7 3 3 8 1 0 3 9 2?

✍ Answers: S6.10 (end of this lecture)

🔁 **RECALL DRILL · ⏱ 4 min · Tier A**

## 7 · Quick recall before you move on

1. Correlation or convolution: which one rotates the kernel by 180° first?
2. A 7×7 image with a 3×3 filter and no padding: what is the output size? And with $p = 1$?
3. Box vs weighted average: which one blurs more, and why?
4. Salt-and-pepper noise: mean filter or median filter? Answer in one sentence.

✍ Answers: S6.11 (end of this lecture)

### ⚠ Traps

- Rotating 180° is *both* flips. Flipping only left–right on a 3×3 kernel gives $\begin{bmatrix} 3 & 7 & 1 \\ 7 & 8 & 8 \\ 5 & 4 & 1 \end{bmatrix}$ for Sheet 6 Q2, which is wrong.
- "Applying a filter" in these slides means correlation (L6 p4). If you convolve, say so: Sheet 6 Q4 gives a different second row.
- No padding mentioned → the output shrinks ($n - k + 1$). Don't silently pad.
- Box blur divides by 9 even when some of the nine are padding zeros; that is why zero-padded borders come out dark.
- The weighted mask 1 2 1 / 2 4 2 / 1 2 1 divides by 16, not 9 (L6 p15 and p17 print $\frac{1}{9}$ by mistake).
- Median: count the padding zeros as values. A corner window has five of them.

## Solutions and answers — Lecture 6

### S6.1 · 2 · Correlation and convolution differ by one 180° turn — Show the full exam answer

**Method (both parts).** Convolution rotates its kernel by 180° before sliding, while correlation does not (L6 p8). So the convolution kernel must be the correlation kernel rotated by 180°: flip it left–right, then up–down. Convolving with the rotated kernel rotates it back, giving the same sum of products at every pixel.

**Q1.** A 1-D kernel only needs the left–right flip: $[1\ 7\ 3] \rightarrow [3\ 7\ 1]$.

Check on any row, say $f = [3\ 1\ 4\ 1\ 5]$ (no padding): correlation with $[1\ 7\ 3]$: $3+7+12=22$, $1+28+3=32$, $4+7+15=26$. Convolution with $[3\ 7\ 1]$ uses $[1\ 7\ 3]$ after its own flip, so it gives the same $[22\ 32\ 26]$.

**Q2.** Flip each row left–right, then reverse the row order:

$$\begin{bmatrix}1&7&3\\8&8&7\\1&4&5\end{bmatrix} \xrightarrow{\text{left–right}} \begin{bmatrix}3&7&1\\7&8&8\\5&4&1\end{bmatrix} \xrightarrow{\text{up–down}} \begin{bmatrix}5&4&1\\7&8&8\\3&7&1\end{bmatrix}$$

**Q1: convolution kernel $[3\ 7\ 1]$. Q2: convolution kernel $\begin{bmatrix}5&4&1\\7&8&8\\3&7&1\end{bmatrix}$. Method: rotate the correlation kernel by 180° (flip horizontally and vertically).**

↩ back to the card

### S6.2 · 2 · Correlation and convolution differ by one 180° turn — Answers

$[6; 5; 4]$: a column flips top to bottom.

↩ back to the card

### S6.3 · 3 · Applying a filter: first the size, then the sums — Show the full exam answer

**Convention.** Filtering = correlation (slides' formula $g = \sum\sum w(s,t)f(x+s,y+t)$, L6 p4), no padding, so the output shrinks to $n-k+1$ per direction.

**Q3.** Size: $5-3+1=3$, so the new sub-image is 1×3.

| window | $3a+4b+5c$ |
|---|---|
| 1 0 1 | $3+0+5=8$ |
| 0 1 0 | $0+4+0=4$ |
| 1 0 1 | $3+0+5=8$ |

**Q3: size 1×3, output $[8\ 4\ 8]$.**

(Convolution gives the same $[8\ 4\ 8]$ here because the sub-image is symmetric. If zero padding were used, the size would stay 1×5 with output $[4\ 8\ 4\ 8\ 4]$.)

**Q4.** Size: rows $4-3+1=2$, columns $5-3+1=3$, so 2×3. Non-zero kernel weights: top row 3, 4, 5; middle-left 1, middle-right 1; bottom-middle 4 (centre is 0).

| output | window (rows × cols) | sum of products |
|---|---|---|

| | | |
|---|---|---|
| (1,1) | 1 0 1 / 0 0 1 / 1 0 1 | $3\cdot1+4\cdot0+5\cdot1+1\cdot0+1\cdot1+4\cdot0=9$ |
| (1,2) | 0 1 0 / 0 1 0 / 0 1 0 | $3\cdot0+4\cdot1+5\cdot0+1\cdot0+1\cdot0+4\cdot1=8$ |
| (1,3) | 1 0 1 / 1 0 0 / 1 0 1 | $3\cdot1+4\cdot0+5\cdot1+1\cdot1+1\cdot0+4\cdot0=9$ |
| (2,1) | 0 0 1 / 1 0 1 / 0 1 1 | $3\cdot0+4\cdot0+5\cdot1+1\cdot1+1\cdot1+4\cdot1=11$ |
| (2,2) | 0 1 0 / 0 1 0 / 1 1 1 | $3\cdot0+4\cdot1+5\cdot0+1\cdot0+1\cdot0+4\cdot1=8$ |
| (2,3) | 1 0 0 / 1 0 1 / 1 1 0 | $3\cdot1+4\cdot0+5\cdot0+1\cdot1+1\cdot1+4\cdot1=9$ |

**Q4: size 2×3, output $\begin{bmatrix}9 & 8 & 9\\11 & 8 & 9\end{bmatrix}$ (correlation, no padding).**

If the examiner means convolution, rotate the filter first to $\begin{bmatrix}0 & 4 & 0\\1 & 0 & 1\\5 & 4 & 3\end{bmatrix}$ and the output becomes $\begin{bmatrix}9 & 8 & 9\\9 & 16 & 11\end{bmatrix}$; the size is 2×3 either way.

↩ back to the card

### S6.4 · 3 · Applying a filter: first the size, then the sums — Answers

$6+4-5+1=6$: the same as the input.

↩ back to the card

### S6.5 · 4 · Padding: give the window something to stand on at the border — Show the full exam answer

**Q5.** Padding per side $p=\frac{k-1}{2}=\frac{3-1}{2}=1$. Add one zero at each end; the length goes from 7 to $7+2=9$:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 1 | 2 | 1 | 2 | 1 | 0 |

Check: filtering this with a 3-long filter gives $9-3+1=7$ outputs, the original length.

**Q5: $p=1$ on each side → [0 1 2 1 2 1 2 1 0] (length 9).**

**Q6.** $p=\frac{3-1}{2}=1$ row/column on every side; the 5×5 image becomes $7\times7$:

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 2 | 1 | 2 | 1 | 0 |
| 0 | 2 | 1 | 2 | 1 | 2 | 0 |
| 0 | 1 | 2 | 1 | 2 | 1 | 0 |
| 0 | 2 | 1 | 2 | 1 | 2 | 0 |
| 0 | 1 | 2 | 1 | 2 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**Q6: $p=1$ → 7×7 array, a ring of zeros around the 5×5 checkerboard; a 3×3 filter then returns a 5×5 output.**

If border (replicate) padding is wanted instead, copy the edge values: Q5 becomes [1 1 2 1 2 1 2 1 1], and Q6's top padded row becomes 1 1 2 1 2 1 1 (the first image row with its end values repeated).

↩ back to the card

### S6.6 · 4 · Padding: give the window something to stand on at the border — Answers

$p = 2$, so $14 \times 14$.

↩ back to the card

### S6.7 · 5 · Box blur by hand, without padding and with zeros — Show the full exam answer

**Filter.** 3×3 averaging (box) mask $\frac{1}{9}\begin{bmatrix}1 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 1\end{bmatrix}$ (L6 p11). Output = window sum ÷ 9, rounded to the nearest integer (the slides' rounding, L6 p13).

**Q3 (no padding).** Output size $(5-3+1) \times (5-3+1) = 3 \times 3$. Column sums per band of three rows:

| rows | column sums | window sums |
|---|---|---|
| 1–3 | 12, 7, 11, 8, 4 | 30, 26, 23 |
| 2–4 | 9, 6, 11, 11, 5 | 26, 28, 27 |
| 3–5 | 8, 7, 10, 9, 8 | 25, 26, 27 |

Sums, then ÷ 9:

| | | |
|---|---|---|
| 30 | 26 | 23 |
| 26 | 28 | 27 |
| 25 | 26 | 27 |

| | | |
|---|---|---|
| 3.33 | 2.89 | 2.56 |
| 2.89 | 3.11 | 3.00 |
| 2.78 | 2.89 | 3.00 |

**Q3:** $\begin{bmatrix}3 & 3 & 3\\ 3 & 3 & 3\\ 3 & 3 & 3\end{bmatrix}$ **(3×3; exact values above if your instructor keeps fractions).**

**Q4 (zero padding).** $p = 1$, the padded array is 7×7 and the output keeps 5×5:

| | | | | | | |
|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 0 | 5 | 2 | 4 | 2 | 1 | 0 |
| 0 | 2 | 1 | 2 | 4 | 2 | 0 |
| 0 | 5 | 4 | 5 | 2 | 1 | 0 |
| 0 | 2 | 1 | 4 | 5 | 2 | 0 |
| 0 | 1 | 2 | 1 | 2 | 5 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| output row | image rows used | column sums (0 at both ends) | window sums |
|---|---|---|---|
| 1 | 1–2 | 0 \| 7, 3, 6, 6, 3 \| 0 | 10, 16, 15, 15, 9 |
| 2 | 1–3 | 0 \| 12, 7, 11, 8, 4 \| 0 | 19, 30, 26, 23, 12 |
| 3 | 2–4 | 0 \| 9, 6, 11, 11, 5 \| 0 | 15, 26, 28, 27, 16 |
| 4 | 3–5 | 0 \| 8, 7, 10, 9, 8 \| 0 | 15, 25, 26, 27, 17 |
| 5 | 4–5 | 0 \| 3, 3, 5, 7, 7 \| 0 | 6, 11, 15, 19, 14 |

÷ 9:

| | | | | |
|---|---|---|---|---|
| 1.11 | 1.78 | 1.67 | 1.67 | 1.00 |
| 2.11 | 3.33 | 2.89 | 2.56 | 1.33 |
| 1.67 | 2.89 | 3.11 | 3.00 | 1.78 |

| | | | | |
|---|---|---|---|---|
| 1.67 | 2.78 | 2.89 | 3.00 | 1.89 |
| 0.67 | 1.22 | 1.67 | 2.11 | 1.56 |

**Q4:** $\begin{bmatrix} 1 & 2 & 2 & 2 & 1 \\ 2 & 3 & 3 & 3 & 1 \\ 2 & 3 & 3 & 3 & 2 \\ 2 & 3 & 3 & 3 & 2 \\ 1 & 1 & 2 & 2 & 2 \end{bmatrix}$ **(5×5). The interior 3×3 equals Q3's answer; only the border changes.**

↩ back to the card

### S6.8 · 5 · Box blur by hand, without padding and with zeros — Answers

The bottom-left corner: $6/9 = 0.67 \rightarrow 1$. Its window holds only four real pixels (2, 1, 1, 2), and they are small.

↩ back to the card

### S6.9 · 6 · The median filter: sort nine values, keep the middle one — Show the full exam answer

**Method.** Pad with one ring of zeros (7×7, see card 5), slide a 3×3 window over every original pixel, sort the 9 values, take the 5th. Sorted windows (median in bold), position = (row, column):

| pixel | sorted window | median |
|---|---|---|
| (1,1) | 0 0 0 0 0 1 2 2 5 | 0 |
| (1,2) | 0 0 0 1 2 2 2 4 5 | 2 |
| (1,3) | 0 0 0 1 2 2 2 4 4 | 2 |
| (1,4) | 0 0 0 1 2 2 2 4 4 | 2 |
| (1,5) | 0 0 0 0 0 1 2 2 4 | 0 |
| (2,1) | 0 0 0 1 2 2 4 5 5 | 2 |
| (2,2) | 1 2 2 2 4 4 5 5 5 | 4 |
| (2,3) | 1 2 2 2 2 4 4 4 5 | 2 |
| (2,4) | 1 1 2 2 2 2 4 4 5 | 2 |
| (2,5) | 0 0 0 1 1 2 2 2 4 | 1 |
| (3,1) | 0 0 0 1 1 2 2 4 5 | 1 |
| (3,2) | 1 1 2 2 2 4 4 5 5 | 2 |
| (3,3) | 1 1 2 2 4 4 4 5 5 | 4 |
| (3,4) | 1 2 2 2 2 4 4 5 5 | 2 |
| (3,5) | 0 0 0 1 2 2 2 4 5 | 2 |
| (4,1) | 0 0 0 1 1 2 2 4 5 | 1 |
| (4,2) | 1 1 1 2 2 4 4 5 5 | 2 |
| (4,3) | 1 1 2 2 2 4 4 5 5 | 2 |
| (4,4) | 1 1 2 2 2 4 5 5 5 | 2 |
| (4,5) | 0 0 0 1 2 2 2 5 5 | 2 |
| (5,1) | 0 0 0 0 0 1 1 2 2 | 0 |
| (5,2) | 0 0 0 1 1 1 2 2 4 | 1 |
| (5,3) | 0 0 0 1 1 2 2 4 5 | 1 |
| (5,4) | 0 0 0 1 2 2 4 5 5 | 2 |
| (5,5) | 0 0 0 0 0 2 2 5 5 | 0 |

**Q5:** $\begin{bmatrix} 0 & 2 & 2 & 2 & 0 \\ 2 & 4 & 2 & 2 & 1 \\ 1 & 2 & 4 & 2 & 2 \\ 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 1 & 2 & 0 \end{bmatrix}$ **(5×5).**

↩ back to the card

### S6.10 · 6 · The median filter: sort nine values, keep the middle one — Answers

Sorted: 0 1 2 3 **3** 3 7 8 9, so the median is 3.

↩ back to the card

### S6.11 · 7 · Quick recall before you move on — Answers

1. Convolution (L6 p8).
2. $7 - 3 + 1 = 5$, so 5×5. With $p = 1$: $7 + 2 - 3 + 1 = 7$, so 7×7.
3. The box filter: its corners count as much as its centre.
4. The median filter: the outlier gets sorted to an end of the list and is never picked, while the mean averages it in and spreads it.

↩ back to the card

LECTURE 7

# Sharpening: derivatives, the Laplacian, unsharp masking, Sobel

Tier A · must ⏱ 53 min

Slides: Lec 7-Sharpening (40 p; p2–p18 sharpening, p20–p39 image features & SIFT) · Sheet 6 Q7–Q13 ↗ Szeliski Ch 3

Lecture 6 blurred images by averaging neighbours. This lecture does the opposite: it measures how fast brightness **changes** from one pixel to the next and adds that change back, so edges get crisper. Every tool here (Laplacian, unsharp mask, Sobel) is still a small mask slid over the image, so the sliding-and-multiplying you learnt in Lecture 6 does all the arithmetic.

**What the exam can ask from this lecture**

- First and second derivative of a 1-D row of pixels (Sheet 6 Q7, Q8), and which one is zero on a ramp.
- Derive the 3×3 Laplacian mask from $\partial^2 f/\partial x^2 + \partial^2 f/\partial y^2$ (Q10) and apply it to a small image (Q9, Q12).
- Sharpen: $g = f - \nabla^2 f$ when the mask centre is negative, $g = f + \nabla^2 f$ when it is positive (L7 p11).
- Unsharp masking and high-boost on a blurred edge (L7 p16).
- Build a 3×3 Gaussian mask from its formula (Q11); low-pass versus high-pass (Q13).
- Sobel gradient of a pixel, magnitude $|G_x| + |G_y|$ (L7 p17–p18).
- Short theory: edge / corner / blob / texture features; SIFT advantages and disadvantages (L7 p25–p39).

⏳ **Only 20 minutes?** Do cards 1, 2 and 4, in that order.

PEN & PAPER · ⏱ 9 min · Tier A

## 1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8)

Read a row left to right: 1, 1, 1, then a jump to 3. A computer notices the jump by **subtracting neighbours**; the whole lecture is built on that.

IN PLAIN WORDS

① A **derivative** measures how much the brightness changes as you step one pixel. Pixels sit on a grid, so there is no limit to take; the slides (L6 p24, repeated in L7 p2) use differences. With $f(x)$ the brightness at position $x$:

$$\frac{\partial f}{\partial x} = f(x+1) - f(x) \qquad \frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

The **first derivative** is "next minus me". The **second derivative** is "my two neighbours minus twice me", which is the change of the change. Q7 calls the first one "Laplacian first derivative"; the Laplacian is really the second-derivative idea, so read Q7 as "first derivative" and Q8 as "second derivative".

WARM-UP BY HAND

② Row $f = [2,\ 2,\ 5,\ 5,\ 3,\ 3]$, positions $x = 0 \ldots 5$.

| x | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| f | 2 | 2 | 5 | 5 | 3 | 3 |
| f′ = f(x+1) − f(x) | 0 | 3 | 0 | −2 | 0 | – |
| f″ = f(x+1)+f(x−1)−2f(x) | – | 3 | −3 | −2 | 2 | – |

THE ONE IDEA

③ The first derivative fires once per jump; the second derivative fires twice, with opposite signs, on the two sides of the jump. Flat stretches give 0 in both. That sign flip (a **zero crossing**) is how the second derivative pinpoints an edge.

BUILD IT

④

1. $f'(1) = 5 - 2 = 3$, $f'(3) = 3 - 5 = -2$, the rest 0.
   Why: "next minus me"; the last pixel has no "next", so 6 pixels give 5 values.
2. $f''(1) = 5 + 2 - 4 = 3$, $f''(2) = 5 + 2 - 10 = -3$, $f''(3) = 3 + 5 - 10 = -2$, $f''(4) = 3 + 5 - 6 = 2$.
   Why: the two end pixels lack a neighbour, so 6 pixels give 4 values.

CHECK

⑤ Add up all the first derivatives: $0 + 3 + 0 - 2 + 0 = 1$, and indeed the row ends at 3 after starting at 2. That telescoping sum is a free check for Q7.

🧭 **Like speed and acceleration in a car.** The first derivative is the speed of the brightness climb; the second is the push you feel only when the speed changes, at the start and end of a climb. *Where it breaks:* pixels jump in whole steps, so there is no smooth limit.

EXAM PROBLEM

**Your turn — Sheet 6 Q7 and Q8.** Q7: "Use Laplacian first derivative in the below array." Q8: "Use laplacian second derivative in the below array."

| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 3 | 3 | 5 | 6 | 6 | 6 | 2 | 1 | 1 | 1 |

✍️ Show the full exam answer: S7.1 (end of this lecture)

TRY ALONE

⑦ Second derivative of the ramp $[1, 2, 3, 4]$?

✍️ Answers: S7.2 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia)

🖍 PEN & PAPER · ⏱ 6 min · Tier A

## 2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10)

An image has rows *and* columns. What if you take card 1's second derivative across the row, again down the column, and add them?

IN PLAIN WORDS

① The **Laplacian** is that sum (L7 p2): $\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$. A **mask** (or kernel) is the little grid of weights you slide over the image; at each pixel you multiply each weight by the pixel under it and add (correlation, Lecture 6). Q10 asks: which 3×3 grid of weights computes $\nabla^2 f$?

**WARM-UP BY HAND**

② One direction only: $f(x-1) - 2f(x) + f(x+1)$ has weights $[1\ -2\ 1]$ on (left, centre, right); the $y$ direction gives the same weights standing up. Patch: centre 7, up 12, left 8, right 11, down 9. Across $8 + 11 - 14 = 5$, down $12 + 9 - 14 = 7$, sum $\nabla^2 f = 12$.

**THE ONE IDEA**

③ Stack the horizontal $[1\ -2\ 1]$ and the vertical $[1\ -2\ 1]$ on one 3×3 grid: the two centres add to $-4$, the four side neighbours get 1, the corners get 0.

**BUILD IT**

④

1. $\frac{\partial^2 f}{\partial x^2} = f(x+1,y) + f(x-1,y) - 2f(x,y)$, $\frac{\partial^2 f}{\partial y^2} = f(x,y+1) + f(x,y-1) - 2f(x,y)$
   Why: card 1's formula, holding $y$ fixed, then $x$.
2. $\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y)$
   Why: add them; the two $-2f(x,y)$ terms merge into $-4f(x,y)$ (L7 p3).
3. Read off the weight of each position: centre $-4$, the four direct neighbours 1, the four diagonals 0 (they never appear).
   Why: a mask is the list of coefficients of the formula, placed where each pixel sits.

**CHECK**

⑤ The weights add to 0, so a flat patch gives 0. Warm-up patch: $12 + 8 + 11 + 9 - 4 \cdot 7 = 12$, as in ②.

Cousins on L7 p9: $-8$ centre with diagonals, and the sign-flipped $+4$ and $+8$. The sign matters for **sharpening**, which adds the Laplacian back to the original (L7 p11):

$$g(x,y) = \begin{cases} f(x,y) - \nabla^2 f(x,y) & \text{centre of the mask negative } (-4,-8) \\ f(x,y) + \nabla^2 f(x,y) & \text{centre of the mask positive } (+4,+8) \end{cases}$$

L7 p12–p13: rows 50, 60, 90, 100 with the $+4$ mask; the "60" pixel gives $240 - 50 - 90 - 60 - 60 = -20$, so $g = 40$. The row becomes 40, 40, 110, 110: a steeper edge.

🧭 **Like the "sharpen" slider in a phone photo editor.** *Where it breaks:* it boosts grain too, since to the Laplacian a speck of TV static looks like a tiny edge.

**EXAM PROBLEM**

**Your turn — Sheet 6 Q10.** "Use Laplacian second derivative to construct 3x3 filter mask."

✍️ Show the full exam answer: S7.3 (end of this lecture)

**TRY ALONE**

⑦ You used the $0, -1, 0 / -1, 4, -1 / 0, -1, 0$ mask. Plus or minus when sharpening?

✍ Answers: S7.4 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia) · ↗ Szeliski Ch 3

🖍 PEN & PAPER · ⏱ 7 min · Tier A

## 3 · Sliding the Laplacian over a small image (Sheet 6 Q9)

Q9's arithmetic is easy; the real decision is **what to do at the border**, where the mask hangs off the picture.

IN PLAIN WORDS

① Place the centre of the $-4$ mask on a pixel, compute up + down + left + right $-$ 4 · centre, move on. Border pixels miss neighbours; L6 p6 offers **ignore the edges** (output shrinks by 1 on each side) or **pad with zeros** (same size, fake edges). Main answer: interior; padded version as a second reading.

WARM-UP BY HAND

② Image $\begin{bmatrix} 2 & 2 & 2 & 2 \\ 5 & 5 & 5 & 5 \\ 5 & 5 & 5 & 5 \end{bmatrix}$: interior is 1×2, both pixels have up 2, down/left/right 5, centre 5, so $17 - 20 = -3$.

THE ONE IDEA

③ The output size is (rows − 2) × (cols − 2) when you ignore the edges; every number is "four neighbours minus four times the centre".

BUILD IT

④

1. Size: $3 \times 4 \to (3-2) \times (4-2) = 1 \times 2$.
   Why: a 3×3 mask needs one pixel of room on every side (Lecture 6's $n - k + 1$).
2. Pixels (1, 1) and (1, 2): $2 + 5 + 5 + 5 - 4 \cdot 5 = -3$.
   Why: the darker row above pulls the sum below $4 \cdot 5$; negative means brighter than the neighbours' average.

CHECK

⑤ Turn the 2s into 5s and you get $20 - 20 = 0$: flat gives 0.

🧭 **Like a magnifying glass sliding over a page:** it rests only where the whole lens fits; at the margins you stop early or tape blank paper around (zero padding). *Where it breaks:* the blank paper is not neutral, the Laplacian sees the jump to 0 as an edge.

EXAM PROBLEM

**Your turn — Sheet 6 Q9.** "Use Laplacian second derivative in the array below."

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 4 | 4 | 4 | 4 | 4 |
| 2 | 4 | 4 | 4 | 2 |

| | | | | |
|---|---|---|---|---|
| 4 | 4 | 4 | 4 | 4 |

✍ Show the full exam answer: S7.5 (end of this lecture)

**TRY ALONE**

⑦ Zero padding: what does the top-left pixel (value 1) of Q9 give?

✍ Answers: S7.6 (end of this lecture)

↗ Kernel (image processing), Wikipedia

🖍 PEN & PAPER · ⏱ 6 min · Tier A

## 4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12)

Q12 is the textbook edge: rows of 50 on rows of 100. Guess first: the Laplacian is non-zero **only on the two rows touching the boundary**.

**IN PLAIN WORDS**

① Apply the 3×3 Laplacian mask (centre −4, card 2) to every interior pixel, then, to show what it is for, sharpen with $g = f - \nabla^2 f$ (negative centre, L7 p11). An **edge** is a place where brightness changes sharply between two regions (L7 p26).

**WARM-UP BY HAND**

② Image with rows 10, 10, 30, 30 (4×4, every row constant). Interior = rows 1–2, columns 1–2.

| row | up | down | left + right | centre | $\nabla^2 f$ | $g = f - \nabla^2 f$ |
|---|---|---|---|---|---|---|
| 1 (value 10) | 10 | 30 | 20 | 10 | 60 - 40 = 20 | 10 - 20 = -10 → clip to 0 |
| 2 (value 30) | 10 | 30 | 60 | 30 | 100 - 120 = -20 | 30 + 20 = 50 |

**THE ONE IDEA**

③ A step gives a +/− pair on the two rows beside it and 0 everywhere else; subtracting it pushes the dark side darker and the bright side brighter. That overshoot is what your eye reads as "sharper".

**BUILD IT**

④

1. Dark row: $10 + 30 + 10 + 10 - 40 = 20$; bright row: $10 + 30 + 30 + 30 - 120 = -20$.
   Why: only the neighbour across the edge differs from the centre.
2. Sharpen: $10 - 20 = -10 \to 0$ (clipped), $30 + 20 = 50$.
   Why: negative centre, so subtract (L7 p11); 8-bit pixels cannot go below 0.

**CHECK**

⑤ Rows far from the edge have all neighbours equal to them, so they give 0.

**EXAM PROBLEM**

**Your turn — Sheet 6 Q12.** "Using the 3x3 Laplacian filter on the below sub-image."

| | | | | | |
|---|---|---|---|---|---|
| 50 | 50 | 50 | 50 | 50 | 50 |
| 50 | 50 | 50 | 50 | 50 | 50 |
| 50 | 50 | 50 | 50 | 50 | 50 |
| 100 | 100 | 100 | 100 | 100 | 100 |
| 100 | 100 | 100 | 100 | 100 | 100 |
| 100 | 100 | 100 | 100 | 100 | 100 |

✍ Show the full exam answer: S7.7 (end of this lecture)

TRY ALONE

⑦ Rows 20, 20, 20, 80, 80, 80: interior Laplacian values, row by row?

✍ Answers: S7.8 (end of this lecture)

↗ Discrete Laplace operator (Wikipedia)

🖍 PEN & PAPER · ⏱ 6 min · Tier A

## 5 · Unsharp masking and high-boost: sharpen by subtracting a blur

An old print-shop trick (L7 p15): to sharpen, make a *blurrier* copy; the difference is **the detail the blur destroyed**, and you add it back.

IN PLAIN WORDS

① $f$ is the original, $\bar{f}$ (written $f'$ on L7 p16) is a blurred copy made with the 3×3 average filter from Lecture 6. The **mask** here is not a kernel but an image: mask $= f - \bar{f}$. Then

$$g_{\text{unsharp}} = f + \text{mask}, \qquad g_{\text{high-boost}} = f + k \cdot \text{mask}, \; k > 1$$

**High-boost** turns the volume of the mask up (the slide uses $k = 3$).

WARM-UP BY HAND

② Row $f = [30, 30, 60, 60]$, ends copied outward: $\bar{f} = [30, 40, 50, 60]$, mask $= [0, -10, 10, 0]$, $g = f + \text{mask} = [30, 20, 70, 60]$, and with $k = 2$: $[30, 10, 80, 60]$.

THE ONE IDEA

③ $f - \bar{f}$ is zero on flat ground and a −/+ pair at an edge; adding it digs a dip before the edge and builds a bump after it.

BUILD IT

④

1. Blur: $\bar{f}(1) = (30 + 30 + 60)/3 = 40$.
   Why: identical rows turn the 3×3 average into a 3-pixel average along the row.
2. Mask $30 - 40 = -10$; $g = 30 - 10 = 20$, or $30 + 2(-10) = 10$ with $k = 2$.
   Why: pushing against the blur undoes it, and a bit more.

CHECK

⑤ The gap across the edge grew from 30 to 50 (unsharp) and 70 ($k = 2$); flat pixels did not move.

**Like an old darkroom trick:** lay an out-of-focus negative over the print and only fine detail survives (hence "unsharp mask"). *Where it breaks:* a big $k$ amplifies noise and leaves bright halos.

EXAM PROBLEM

**Your turn — L7 p16.** "Apply Unsharp filter on the following blurred edge." Every row of $f$ is $[50,\ 60,\ 90,\ 100]$ (4×4). Blur with the 3×3 standard average (all weights $1/9$), then give $g_{\text{unsharp}} = f + \text{mask}$ and $g_{\text{highboost}} = f + 3 \cdot \text{mask}$.

Show the full exam answer: S7.9 (end of this lecture)

TRY ALONE

⑦ Unsharp the row $[30, 30, 60, 60]$ with $k = 3$.

Answers: S7.10 (end of this lecture)

↗ Unsharp masking (Wikipedia)

PEN & PAPER · ⏱ 5 min · Tier A

## 6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11)

A box average gives every neighbour the same vote; a **Gaussian** mask lets near pixels count more, following the bell curve.

IN PLAIN WORDS

① $h(x, y) = \dfrac{1}{2\pi\sigma^2}\, e^{-\frac{x^2+y^2}{2\sigma^2}}$, where $(x, y)$ is the position *relative to the mask centre* (so $x, y \in \{-1, 0, 1\}$) and $\sigma$ is the **standard deviation**, the width of the bell (the sheet writes it as $\alpha$). Only $x^2 + y^2$ matters, and on a 3×3 grid it takes three values: 0 (centre), 1 (the four side cells), 2 (the four corners).

WARM-UP BY HAND

② Same recipe with $\sigma = 2$: $2\sigma^2 = 8$ and $\frac{1}{2\pi\sigma^2} = \frac{1}{8\pi} = 0.0398$.

| cell | $x^2 + y^2$ | $h$ | normalised ($\div 0.3042$) |
|---|---|---|---|
| centre | 0 | $0.0398 \cdot e^{0} = 0.0398$ | **0.1308** |
| side (×4) | 1 | $0.0398 \cdot e^{-1/8} = 0.0351$ | **0.1154** |
| corner (×4) | 2 | $0.0398 \cdot e^{-1/4} = 0.0310$ | **0.1019** |

THE ONE IDEA

③ Compute three numbers (centre, side, corner), place them symmetrically, then divide by their total so the nine weights add to 1. Without that last step a blur would darken the image (weights summing to 0.30 keep only 30% of the brightness).

BUILD IT

④

1. $h = 0.0398$ (centre), $0.0398 \cdot e^{-1/8} = 0.0351$ (side), $0.0398 \cdot e^{-2/8} = 0.0310$ (corner).

Why: $x^2 + y^2$ is 0, 1, 2 for centre, side, corner.

2. Total $0.0398 + 4(0.0351) + 4(0.0310) = 0.3042$; divide each weight by it.
   Why: weights summing to 1 keep flat areas at their brightness.

**CHECK**

⑤ $0.1308 + 4(0.1154) + 4(0.1019) = 1.000$.

🧭 **Like Instagram's soft-focus blur** versus a blocky box blur: influence fades smoothly with distance. *Where it breaks:* the true bell never reaches zero; a 3×3 cut drops its tails, hence the renormalising.

**EXAM PROBLEM**

**Your turn — Sheet 6 Q11.** "Using the gaussian function below to construct 3x3 filter mask. $h(x, y) = \frac{1}{2\pi\alpha^2} e^{-\frac{x^2+y^2}{2\alpha^2}}$, and $\alpha$ which is standard deviation equals 1."

✍ Show the full exam answer: S7.11 (end of this lecture)

**TRY ALONE**

⑦ If $\sigma$ shrinks to 0.5, does the normalised centre weight go up or down?

✍ Answers: S7.12 (end of this lecture)

↗ Gaussian blur (Wikipedia)

📖 READ & UNDERSTAND · ⏱ 3 min · Tier A

## 7 · Low-pass versus high-pass: the two families side by side (Sheet 6 Q13)

Think of an image as slow changes (large smooth areas, the sky) plus fast changes (edges, fine texture, noise). **Frequency** here means how quickly brightness changes from pixel to pixel. A **low-pass filter** lets the slow changes through and removes the fast ones: it smooths (L6 p10). A **high-pass filter** does the reverse: it keeps the fast changes and removes the flat background, so edges and detail stand out.

The weight sum gives it away. Average and Gaussian masks have positive weights **summing to 1**: a flat area of 80 stays 80. The Laplacian and Sobel masks have weights **summing to 0**: a flat area of 80 becomes $80 \cdot 0 = 0$, only change survives. Tiny number: the row $[10, 10, 40, 40]$ under the average $[1, 1, 1]/3$ gives 20 and 30 in the middle (the jump of 30 is softened to 10 per step), while the second derivative gives $+30$ and $-30$ there (only the jump shows).

🧭 **Like the bass and treble knobs on a speaker.** Low-pass keeps the bass (smooth shading), high-pass keeps the treble (crisp edges, and also the hiss). *Where it breaks:* images are 2-D, so "fast change" can point in any direction, which is why Sobel needs two masks.

**EXAM PROBLEM**

**Your turn — Sheet 6 Q13.** "What is the difference between low-pass filter and high-pass filter."

✍ Show the full exam answer: S7.13 (end of this lecture)

↗ Kernel (image processing), Wikipedia

PEN & PAPER · ⏱ 6 min · Tier A

## 8 · The Sobel gradient: how strong is the edge at this pixel?

The Laplacian ignores direction; Sobel measures change **along each axis separately**, then combines them.

IN PLAIN WORDS

① The **gradient** is the pair (change left-to-right, change top-to-bottom). Sobel computes each with a 3×3 mask (L7 p17). Using the slide's labels:

| $G_x$ (right column − left column) | | | | $G_y$ (bottom row − top row) | | |
|---|---|---|---|---|---|---|
| -1 | 0 | 1 | | -1 | -2 | -1 |
| -2 | 0 | 2 | | 0 | 0 | 0 |
| -1 | 0 | 1 | | 1 | 2 | 1 |

Edge strength (**magnitude**) on the slide: $|\nabla f| \approx |G_x| + |G_y|$ (L7 p18), a cheap stand-in for $\sqrt{G_x^2 + G_y^2}$.

WARM-UP BY HAND

② Neighbourhood $\begin{bmatrix} 10 & 10 & 50 \\ 10 & 10 & 50 \\ 10 & 10 & 50 \end{bmatrix}$, a vertical edge: $G_x = (50-10) + 2(50-10) + (50-10) = 160$, $G_y = 80 - 80 = 0$, so $|G| \approx 160$.

THE ONE IDEA

③ Each Sobel mask is "one side minus the other side" with weights 1, 2, 1; a flat neighbourhood gives 0, an edge across the mask gives a big number. $G_x$ reacts to vertical edges (brightness changes as you go right), $G_y$ to horizontal ones.

BUILD IT

④

1. $G_x$: right column $(50, 50, 50) \cdot (1, 2, 1) = 200$, left column $-40$; total 160.
   Why: correlation, weight times pixel, then add.
2. $G_y = 80 - 80 = 0$; $|G| \approx 160 + 0$.
   Why: identical rows, nothing changes going down.

CHECK

⑤ All 10s gives 0 for both masks: a constant area has zero gradient (L7 p18). Not on the slides: **Prewitt** uses 1, 1, 1 instead of 1, 2, 1; **Roberts** uses 2×2 diagonal differences.

🧭 **Like a spirit level laid twice,** east–west and north–south; together they give the steepness. *Where it breaks:* $|G_x| + |G_y|$ overstates diagonal edges by up to about 1.4×.

EXAM PROBLEM

**Your turn — L7 p18.** "Find the gradient of the following shaded pixels" with the Sobel masks above and $V = |G_x| + |G_y|$. Image (4×4):

| | | | |
|---|---|---|---|
| 10 | 10 | 200 | 200 |
| 10 | 10 | 200 | 200 |
| 100 | 100 | 20 | 20 |
| 100 | 100 | 20 | 20 |

The slide asks for pixel (1,1); find it, then the other three interior pixels.

✍ Show the full exam answer: S7.14 (end of this lecture)

**TRY ALONE**

⑦ $G_y$ for $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 9 & 9 & 9 \end{bmatrix}$?

✍ Answers: S7.15 (end of this lecture)

↗ Sobel operator (Wikipedia) · ↗ Prewitt · ↗ Roberts cross

📖 READ & UNDERSTAND · ⏱ 2 min · Tier C

## 9 · Image features and SIFT, in one paragraph

Edges are one of four kinds of **image features**, the distinctive parts a computer uses to categorise and match pictures (L7 p25–p29): **edges** (boundaries where intensity changes sharply), **corners** (where two or more edges meet), **blobs** (regions of similar intensity or colour, often round objects) and **texture** (a repeating pattern). **SIFT** (L7 p35–p39) finds keypoints and describes them by local gradient directions. Advantages: invariant to scale and rotation, distinctive, robust to noise and partial occlusion. Disadvantages: computationally expensive (scale-space pyramid, keypoints, descriptors) and memory-hungry. Tiny number: the corner patch /usr/bin/bash,0,0 / 0,9,9 / 0,9,9$gives Sobel = G_y = 27$, strong in both directions, while an edge is strong in one.

🧭 **Like recognising a friend by their face, not their pixel position.** *Where it breaks:* a blank wall has no keypoints to match.

↗ OpenCV SIFT tutorial (linked on L7 p35)

🔁 RECALL DRILL · ⏱ 3 min · Tier A

## 10 · Quick recall before the exam

1. Write the first and the second derivative of $f(x)$ as differences.
2. Second derivative of $[5, 5, 9, 9]$ at its two interior points?
3. You sharpen with the mask whose centre is $-8$. Is it $g = f + \nabla^2 f$ or $g = f - \nabla^2 f$?
4. Unsharp masking: write the mask and the high-boost formula.
5. Sum of weights of a Gaussian mask after normalising? Of a Laplacian mask?
6. Sobel gives $G_x = -30$, $G_y = 40$. Edge strength by the slides' rule?

✍ Answers: S7.16 (end of this lecture)

### ⚠ Traps

- Sheet 6 Q7 says "Laplacian first derivative". There is no such thing: compute the plain first derivative $f(x+1) - f(x)$ and say so in one line.
- Count your outputs: $n$ pixels give $n-1$ first-derivative values and $n-2$ second-derivative values; a 3×3 mask with edges ignored shrinks each side by 2.
- The sharpening sign follows the mask centre: negative centre ($-4$, $-8$) means subtract, positive centre means add. Mixing them blurs instead of sharpening.
- Gaussian: $(x, y)$ are offsets from the centre ($-1, 0, 1$), not pixel indices 0–2; and normalise so the weights add to 1.
- Slide L7 p18 says the gradient at (1,1) is 0 "because it is in a constant area". In the 4×4 image shown, the interior pixel (1,1) is next to the 200s and 100s and has $|G_x| + |G_y| = 580$; the 0 only holds for the top-left corner with copied borders. State which reading you use. The same slide also swaps the $G_x/G_y$ names between figure and arithmetic.
- Sharpened values can fall below 0 or rise above 255: clip them (card 4's warm-up).

## Solutions and answers — Lecture 7

### S7.1 · 1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8) — Show the full exam answer

**Q7 — first derivative**, $f'(x) = f(x+1) - f(x)$ (L6 p24), for $x = 0 \ldots 11$ (13 pixels give 12 values). The sheet says "Laplacian first derivative"; the first derivative is meant.

| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f | 1 | 1 | 1 | 3 | 3 | 5 | 6 | 6 | 6 | 2 | 1 | 1 | 1 |
| f′ | 0 | 0 | 2 | 0 | 2 | 1 | 0 | 0 | -4 | -1 | 0 | 0 | – |

Working: $1-1=0, 1-1=0, 3-1=2, 3-3=0, 5-3=2, 6-5=1, 6-6=0, 6-6=0, 2-6=-4, 1-2=-1, 1-1=0, 1-1=0$. Check: sum $= 0$, and the row starts and ends at 1.

$f' = [0,\ 0,\ 2,\ 0,\ 2,\ 1,\ 0,\ 0,\ -4,\ -1,\ 0,\ 0]$

**Q8 — second derivative**, $f''(x) = f(x+1) + f(x-1) - 2f(x)$, for the interior $x = 1 \ldots 11$ (11 values).

| x | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| f | 1 | 1 | 1 | 3 | 3 | 5 | 6 | 6 | 6 | 2 | 1 | 1 | 1 |
| f″ | – | 0 | 2 | -2 | 2 | -1 | -1 | 0 | -4 | 3 | 1 | 0 | – |

Working: $x$=1: $1+1-2=0$; $x$=2: $3+1-2=2$; $x$=3: $3+1-6=-2$; $x$=4: $5+3-6=2$; $x$=5: $6+3-10=-1$; $x$=6: $6+5-12=-1$; $x$=7: $6+6-12=0$; $x$=8: $2+6-12=-4$; $x$=9: $1+6-4=3$; $x$=10: $1+2-2=1$; $x$=11: $1+1-2=0$.

$f'' = [0,\ 2,\ -2,\ 2,\ -1,\ -1,\ 0,\ -4,\ 3,\ 1,\ 0]$ **(for** $x = 1 \ldots 11$**)**

Reading it: both derivatives are 0 on the flat parts (1 1 1 and 6 6 6). Each jump gives one first-derivative spike but a +/− pair in the second derivative (2, −2 around the step 1→3; −4, 3 around the drop 6→2): the zero crossings mark the edges. Some books also write the first derivative as $f(x) - f(x-1)$; the numbers are the same, shifted one place right.

↩ back to the card

### S7.2 · 1 · Derivatives of a row of pixels (Sheet 6 Q7, Q8) — Answers

$[0, 0]$: on a steady ramp the second derivative is 0 (L6 p26), while the first is a constant 1.

↩ back to the card

### S7.3 · 2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10) — Show the full exam answer

Laplacian: $\nabla^2 f = \dfrac{\partial^2 f}{\partial x^2} + \dfrac{\partial^2 f}{\partial y^2}$.

Second derivative in $x$: $\dfrac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$.

Second derivative in $y$: $\dfrac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$.

Adding: $\nabla^2 f = [f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1)] - 4f(x,y)$.

Placing each coefficient at its pixel position (centre $= (x,y)$):

| | | |
|---|---|---|
| 0 | 1 | 0 |
| 1 | -4 | 1 |
| 0 | 1 | 0 |

**Laplacian mask** $= \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ **(weights sum to 0)**

Worth adding for full marks: if the diagonal second derivatives are included too, the mask is $\begin{bmatrix} 1 & 1 & 1 \\ 1 & -8 & 1 \\ 1 & 1 & 1 \end{bmatrix}$; multiplying by $-1$ gives the equivalent masks with centre $+4$ or $+8$ (L7 p9). To sharpen, use $g = f - \nabla^2 f$ for a negative centre and $g = f + \nabla^2 f$ for a positive centre (L7 p11).

↩ back to the card

### S7.4 · 2 · From the second derivative to the 3×3 Laplacian mask (Sheet 6 Q10) — Answers

Plus: the centre is positive, so $g = f + \nabla^2 f$.

↩ back to the card

### S7.5 · 3 · Sliding the Laplacian over a small image (Sheet 6 Q9) — Show the full exam answer

Mask (from Q10): $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, so $\nabla^2 f(x,y) = f(\text{up}) + f(\text{down}) + f(\text{left}) + f(\text{right}) - 4f(x,y)$. Edges ignored (L6 p6): a 4×5 image gives a $(4-2) \times (5-2) = 2 \times 3$ output, for rows 1–2 and columns 1–3 (counting from 0).

| pixel | up | down | left | right | centre | result |
|---|---|---|---|---|---|---|
| (1,1) | 1 | 4 | 4 | 4 | 4 | 13 - 16 = -3 |
| (1,2) | 1 | 4 | 4 | 4 | 4 | 13 - 16 = -3 |
| (1,3) | 1 | 4 | 4 | 4 | 4 | 13 - 16 = -3 |
| (2,1) | 4 | 4 | 2 | 4 | 4 | 14 - 16 = -2 |
| (2,2) | 4 | 4 | 4 | 4 | 4 | 16 - 16 = 0 |
| (2,3) | 4 | 4 | 4 | 2 | 4 | 14 - 16 = -2 |

$\nabla^2 f = \begin{bmatrix} -3 & -3 & -3 \\ -2 & 0 & -2 \end{bmatrix}$ **(2×3, edges ignored)**

Second reading, zero padding (same 4×5 size; the border values are artefacts of the fake 0 frame):

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 2 | 2 | 1 |
| -9 | -3 | -3 | -3 | -9 |
| 4 | -2 | 0 | -2 | 4 |
| -10 | -4 | -4 | -4 | -10 |

The interior 2×3 block is the same in both readings. If the $-8$ mask (with diagonals) is used instead, the interior is $\begin{bmatrix} -11 & -9 & -11 \\ -2 & 0 & -2 \end{bmatrix}$. With the $+4$ mask every sign flips.

↩ back to the card

### S7.6 · 3 · Sliding the Laplacian over a small image (Sheet 6 Q9) — Answers

Up 0, left 0, right 1, down 4: $5 - 4 = 1$.

↩ back to the card

### S7.7 · 4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12) — Show the full exam answer

Mask $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$, edges ignored, so the 6×6 image gives a 4×4 output (rows 1–4, columns 1–4). Each row of the image is constant, so left + right = 2·centre and only up/down matter.

| image row | up | down | centre | $\nabla^2 f$ |
|---|---|---|---|---|
| 1 (50) | 50 | 50 | 50 | 50+50+50+50 - 200 = 0 |
| 2 (50) | 50 | 100 | 50 | 50+100+50+50 - 200 = 50 |
| 3 (100) | 50 | 100 | 100 | 50+100+100+100 - 400 = -50 |
| 4 (100) | 100 | 100 | 100 | 400 - 400 = 0 |

$$\nabla^2 f = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 50 & 50 & 50 & 50 \\ -50 & -50 & -50 & -50 \\ 0 & 0 & 0 & 0 \end{bmatrix} \textbf{ (4×4, edges ignored)}$$

Sharpened interior, $g = f - \nabla^2 f$ (negative centre):

| | | | |
|---|---|---|---|
| 50 | 50 | 50 | 50 |
| 0 | 0 | 0 | 0 |
| 150 | 150 | 150 | 150 |
| 100 | 100 | 100 | 100 |

The edge between 50 and 100 now jumps from 0 to 150: much stronger contrast. With the $+4$ mask the Laplacian rows are $0, -50, 50, 0$ and you use $g = f + \nabla^2 f$, giving the same $g$. Full-size versions: replicate padding (the slides' example style) gives 6×6 with rows $0, 0, 50, -50, 0, 0$; zero padding gives fake values on the border (top-left corner $50 + 50 - 200 = -100$), the artefact L6 p6 warns about.

↩ back to the card

### S7.8 · 4 · A clean step edge, filtered and then sharpened (Sheet 6 Q12) — Answers

$0, 60, -60, 0$; sharpening changes only the two edge rows ($-40 \to 0$ and 140).

↩ back to the card

### S7.9 · 5 · Unsharp masking and high-boost: sharpen by subtracting a blur — Show the full exam answer

Convention (from the slide's numbers): border pixels are repeated outward, and the blurred values are rounded to the nearest integer. Because all four rows are equal, the 3×3 average of each pixel equals the

average of three neighbours along the row.

| | x=0 | x=1 | x=2 | x=3 |
|---|---|---|---|---|
| f | 50 | 60 | 90 | 100 |
| $\bar{f}$ | (50+50+60)/3 = 53.3 → 53 | (50+60+90)/3 = 66.7 → 67 | (60+90+100)/3 = 83.3 → 83 | (90+100+100)/3 = 96.7 → 97 |
| mask = f − $\bar{f}$ | −3 | −7 | 7 | 3 |
| f + mask | 47 | 53 | 97 | 103 |
| f + 3·mask | 50 − 9 = 41 | 60 − 21 = 39 | 90 + 21 = 111 | 100 + 9 = 109 |

**Every row:** $\bar{f} = [53, 67, 83, 97]$, **mask** $= [-3, -7, 7, 3]$, $g_{\text{unsharp}} = [47, 53, 97, 103]$, $g_{\text{highboost}} = [41, 39, 111, 109]$

Comment (as on the slide): the edge pair moved from $(60, 90)$, a gap of $30$, to $(53, 97)$, a gap of $44$, and with high-boost to $(39, 111)$, a gap of $72$. If unsharp masking is not enough, multiply the mask by $k > 1$.

↩ back to the card

### S7.10 · 5 · Unsharp masking and high-boost: sharpen by subtracting a blur — Answers

$[30,\ 0,\ 90,\ 60]$.

↩ back to the card

### S7.11 · 6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11) — Show the full exam answer

$\sigma = \alpha = 1$: $2\sigma^2 = 2$, $\frac{1}{2\pi\sigma^2} = \frac{1}{2\pi} = 0.1592$. Mask positions $x, y \in \{-1, 0, 1\}$.

- Centre $(0, 0)$: $h = 0.1592 \cdot e^{0} = 0.1592$
- Sides $(\pm 1, 0), (0, \pm 1)$: $h = 0.1592 \cdot e^{-1/2} = 0.1592 \cdot 0.6065 = 0.0965$
- Corners $(\pm 1, \pm 1)$: $h = 0.1592 \cdot e^{-2/2} = 0.1592 \cdot 0.3679 = 0.0585$

Raw mask:

| | | |
|---|---|---|
| 0.0585 | 0.0965 | 0.0585 |
| 0.0965 | 0.1592 | 0.0965 |
| 0.0585 | 0.0965 | 0.0585 |

Sum $= 0.1592 + 4(0.0965) + 4(0.0585) = 0.7795$ (unrounded; with the 4-decimal values you get 0.7792, and either is accepted). Normalised mask = raw ÷0.7795:

| | | |
|---|---|---|
| 0.0751 | 0.1238 | 0.0751 |
| 0.1238 | 0.2042 | 0.1238 |
| 0.0751 | 0.1238 | 0.0751 |

**Gaussian 3×3, $\sigma = 1$: centre 0.2042, sides 0.1238, corners 0.0751 (raw: 0.1592, 0.0965, 0.0585); weights sum to 1**

↩ back to the card

### S7.12 · 6 · Building a 3×3 Gaussian mask from its formula (Sheet 6 Q11) — Answers

Up: a narrow bell piles almost all the weight on the centre.

↩ back to the card

### S7.13 · 7 · Low-pass versus high-pass: the two families side by side (Sheet 6 Q13) — Show the full exam answer

| | Low-pass (smoothing) filter | High-pass (sharpening) filter |
|---|---|---|
| Passes | low frequencies: slowly varying areas | high frequencies: edges, fine detail, noise |
| Removes | noise, small details, sharp transitions | constant / slowly varying background |
| Effect | blurs the image, reduces noise | highlights edges, sharpens (after adding back to $f$) |
| Based on | averaging (integration) | differences (derivatives) |
| Mask weights | positive, sum to 1 | positive and negative, sum to 0 |
| Examples | box average $\frac{1}{9}[1\dots1]$, weighted average, Gaussian (median is a non-linear smoother) | Laplacian, Sobel, unsharp mask $f - \bar{f}$ |

**Low-pass keeps slow intensity changes and blurs / removes noise; high-pass keeps fast changes (edges) and removes flat regions. Low-pass weights sum to 1, high-pass weights sum to 0.**

↩ back to the card

### S7.14 · 8 · The Sobel gradient: how strong is the edge at this pixel? — Show the full exam answer

**What the slide claims.** It writes $G_x(1,1) = -10-20-10+0+0+0+10+20+10 = 0$, $G_y(1,1) = 0$, $V(1,1) = 0$, "because it is located within a constant area". That only holds if $(1,1)$ is the top-left pixel (counting from 1) and the missing border is filled by copying edge pixels: then its 3×3 window is all 10s. As a real interior pixel (row 1, column 1, counting from 0, the second 10) its window is $\begin{bmatrix} 10 & 10 & 200 \\ 10 & 10 & 200 \\ 100 & 100 & 20 \end{bmatrix}$, which is not constant.

Interior pixel (1,1), counting from 0:

- $G_x = (200-10) + 2(200-10) + (20-100) = 190 + 380 - 80 = 490$
- $G_y = (100+200+20) - (10+20+200) = 320 - 230 = 90$
- $V = |490| + |90| = 580$

The other interior pixels, same method (window = the 3×3 block centred on the pixel):

| pixel | window | $G_x$ | $G_y$ | $V$ |
|---|---|---|---|---|
| (1,1) | 10 10 200 / 10 10 200 / 100 100 20 | 490 | 90 | 580 |
| (1,2) | 10 200 200 / 10 200 200 / 100 20 20 | 490 | -450 | 940 |
| (2,1) | 10 10 200 / 100 100 20 / 100 100 20 | -50 | 90 | 140 |
| (2,2) | 10 200 200 / 100 20 20 / 100 20 20 | -50 | -450 | 500 |

**Top-left pixel with replicated borders (the slide's reading):** $V = 0$. **Interior pixels:** $V = \begin{bmatrix} 580 & 940 \\ 140 & 500 \end{bmatrix}$

Labels: on p17 the mask with rows $-1\ -2\ -1 / 0\,0\,0 / 1\,2\,1$ is called $G_y$, but the p18 arithmetic for "$G_x$" uses exactly those weights. The names are swapped between figure and arithmetic; $V$ is the same either way.

↩ back to the card

### S7.15 · 8 · The Sobel gradient: how strong is the edge at this pixel? — Answers

$9 + 18 + 9 = 36$ (and $G_x = 0$).

↩ back to the card

### S7.16 · 10 · Quick recall before the exam — Answers

1. $f'(x) = f(x+1) - f(x)$; $f''(x) = f(x+1) + f(x-1) - 2f(x)$.
2. $9 + 5 - 10 = 4$ and $9 + 5 - 18 = -4$: a +/− pair around the step.
3. $g = f - \nabla^2 f$ (negative centre, L7 p11).
4. mask $= f - \bar{f}$ (original minus blurred); $g = f + k \cdot$ mask with $k > 1$ ($k = 1$ is plain unsharp masking).
5. 1 (low-pass, keeps brightness); 0 (high-pass, flat areas vanish).
6. $|-30| + |40| = 70$ (the exact magnitude $\sqrt{900 + 1600} = 50$).

↩ back to the card

# The formula sheet — write it from memory before the exam

Cover the right column, say each line out loud, then check. Star anything you missed and redo the starred lines once more.

| | |
|---|---|
| **Lectures 2–3 · digital images, pixels, operations** | |
| Levels and bits | $L = 2^k$ (8 bits → 256 levels, 3 bits → 8) |
| Image size | $M \times N \times$ channels $\times\ k$ bits; ÷ 8 = bytes; ÷ $2^{20}$ = MB |
| Transmission | bits on the line = pixels × (data bits + start + stop) per packet; time = bits ÷ baud |
| Neighbours | $N_4$: $(x \pm 1, y), (x, y \pm 1)$ · $N_D$: $(x \pm 1, y \pm 1)$ · $N_8 = N_4 \cup N_D$ |
| m-adjacency (diagonal pair, both in $V$) | allowed iff $N_4(p) \cap N_4(q)$ has no pixel with value in $V$ |
| Distances | $D_e = \sqrt{(x-s)^2 + (y-t)^2}$ · $D_4 = \lvert x-s\rvert + \lvert y-t\rvert$ · $D_8 = \max(\lvert x-s\rvert, \lvert y-t\rvert)$ |
| Bilinear interpolation | $v(x,y) = ax + by + cxy + d$ from the 4 nearest pixels; bicubic uses 16 |
| Linear operator | $H[a_1 f_1 + a_2 f_2] = a_1 H[f_1] + a_2 H[f_2]$; one counter-example disproves it |
| ROI / noise | ROI $= f \odot$ mask (array product) · corrupted $= f + \eta$, clip to 0–255 |
| **Lectures 4–5 · intensity transformations and histograms** | |
| Negative · log · gamma | $s = L - 1 - r$ · $s = c\log(1 + r)$ · $s = c\,r^{\gamma}$ |
| Contrast stretching | $I_{new} = (I - I_{min})\dfrac{new_{max} - new_{min}}{I_{max} - I_{min}} + new_{min}$ |
| Histogram | $h(r_k) = n_k$, $p(r_k) = n_k / MN$ |
| Equalization | $s_k = (L-1)\sum_{j=0}^{k} p_r(r_j)$, rounded to the nearest level |
| Matching (specification) | equalize the input → $s_k$; $G(z_q) = (L-1)\sum_{i=0}^{q} p_z(z_i)$, rounded; map each $s_k$ to the $z_q$ whose $G(z_q)$ is closest |
| **Lecture 6 · spatial filtering** | |
| Correlation | $g(x,y) = \sum_s \sum_t w(s,t)\, f(x+s, y+t)$ |
| Convolution | correlation with the kernel rotated 180°; to make a convolution give a correlation's output, rotate the kernel 180° first |
| Output size and padding | no padding: $n - k + 1$ · padding $p = (k-1)/2$ on each side keeps size $n$ · full: $n + k - 1$ |
| Smoothing | box: $\frac{1}{9}$ of all-ones $3 \times 3$ · weighted: $\frac{1}{16}\begin{bmatrix}1&2&1\\2&4&2\\1&2&1\end{bmatrix}$ · median: middle of the 9 sorted values |
| **Lecture 7 · sharpening** | |
| First derivative | $\dfrac{\partial f}{\partial x} = f(x+1) - f(x)$ |
| Second derivative | $\dfrac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$ |

| | |
|---|---|
| Laplacian | $\nabla^2 f = f(x+1,y) + f(x-1,y) + f(x,y+1) + f(x,y-1) - 4f(x,y) \rightarrow$ mask $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ (with diagonals: centre $-8$) |
| Sharpening | $g = f + c\,\nabla^2 f$, with $c = -1$ when the mask's centre is negative |
| Gaussian, $\sigma = 1$ | $h(x,y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}}$: centre 0.1592, edge 0.0965, corner 0.0585; divide by their sum to normalise |
| Unsharp masking / high-boost | $g_{mask} = f - \bar{f}$ (blurred); $g = f + k\,g_{mask}$, $k = 1$ unsharp, $k > 1$ high-boost |
| Low-pass vs high-pass | low-pass (smoothing) keeps slow changes, removes detail and noise; high-pass (sharpening) keeps edges and fine detail |